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**An Explainable AI Framework for Multi-Source Big Data in SME Strategic
Decision-Making**

von

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List of Abbreviation

AI - Artificial Intelligence

BDA – Big Data Analytics

DSS – Decision Support System

EDA – Exploratory Data Analysis

HR – Human Resources

KPI – Key Performance Indicator

LIME – Local Interpretable Model-agnostic Explanations

MAE – Mean Absolute Error

ML – Machine Learning

RF – Random Forest

RMSE – Root Mean Squared Error

ROC-AUC – Receiver Operating Characteristic – Area Under the Curve

SHAP – SHapley Additive exPlanations

SME – Small and Medium-Sized Enterprise

XAI – Explainable Artificial Intelligence

XGBoost – Extreme Gradient Boosting

Abstract

The increasing availability of Artificial Intelligence (AI) presents significant opportunities for Small and Medium-Sized Enterprises (SMEs) to enhance strategic decision-making, operational efficiency, and competitive positioning. Despite this potential, AI adoption within SMEs remains limited due to resource constraints, limited analytical expertise, and the lack of transparency inherent in many machine learning models. The resulting opacity of AI-driven predictions reduces managerial trust and restricts their practical use in strategic decision contexts.

This thesis proposes and evaluates an explainable AI framework designed to support strategic decision-making in SMEs through the integration of multi-source big data, predictive analytics, and model interpretability. Using a realistically constructed synthetic dataset representing key SME operational domains, machine learning models were developed for both regression and classification tasks relevant to business strategy. Explainability was operationalised using Local Interpretable Model-agnostic Explanations (LIME) to provide instance-level insights into model predictions.

The results demonstrate that the integration of multi-source data improves predictive relevance, while LIME enables complex model outputs to be translated into clear, intuitive explanations aligned with managerial reasoning. By enhancing transparency and supporting informed judgement, the proposed framework improves trust in AI-driven decision support without requiring advanced technical expertise. This study contributes a practical and scalable approach to responsible AI adoption, demonstrating how explainable AI can bridge the gap between predictive performance and real-world usability in SME strategic decision-making.

Keywords: Explainable Artificial Intelligence, XAI, SME, LIME, Small and Medium-Sized Enterprises, Strategic Decision-Making, Machine Learning, Multi-Source Big Data, Decision Support Systems

1. Introduction

1.1 Background

1.1.1 Evolution of Artificial Intelligence in Business Decision-Making

In the last few years, the way businesses use artificial intelligence (AI) has changed a lot. Most of the time, big companies with a lot of data and processing power were the first to use AI systems. But now that cloud computing and big data technology are here, and machine learning techniques are growing quickly, AI is easier for businesses of all sizes to use, even small and medium-sized ones.¹

Artificial intelligence (AI), especially machine learning (ML) algorithms, has changed how people make decisions. These algorithms let businesses process huge amounts of data, find patterns, and make predictions. These features help businesses make customers happier, make processes more efficient, and even predict market trends.² AI-driven solutions are now necessary for keeping businesses competitive in the market, increasing productivity, and lowering costs in fields like manufacturing, retail, and finance.

But even though AI has a lot of potential, many small and medium-sized businesses still have trouble using it. This is mostly because machine learning models are hard to understand and complicated. Because AI-driven predictions are so complicated, people who don't work in technology find it hard to believe them and act on them. If AI systems don't make their decision-making process clear, SME leaders might not want to fully use these technologies.³

1.1.2 The Role of Explainable AI (XAI) in Overcoming Barriers

Explainable AI (XAI) wants to fix these problems by making AI decisions clearer and easier to understand. XAI methods aim to make AI systems easier to understand so that

¹ Cf. Chen, H., Chiang, R. H. L. & Storey, V. C., Business Intelligence and Analytics: From Big Data to Big Impact, *MIS Quarterly*, Vol. 36, No. 4, 2012, pp. 1165–1188.

² Cf. Massaro, M., Big Data and Decision-Making: A Structured Literature Review, *Management Decision*, Vol. 61, No. 3, 2023, pp. 653–671.

³ Cf. Vilone, G. & Longo, L., Explainable Artificial Intelligence: A Systematic Review, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

users can not only get predictions but also understand why they are getting them. This is important for building trust in AI systems, especially in businesses where the people making decisions aren't tech-savvy.⁴

XAI techniques are very important in fields like healthcare, finance, and small businesses, where AI-driven decisions can have big ethical, legal, and financial effects.⁵ XAI can help small and medium-sized businesses (SMEs) get the most out of AI and make better, data-driven decisions by making sure that decision-makers understand how AI models come to their conclusions.

XAI uses a lot of a method called Local Interpretable Model-agnostic Explanations (LIME). LIME lets users see which features had the biggest effect on a choice by giving them local explanations for each prediction. LIME helps small and medium-sized businesses turn complicated, hard-to-understand models into useful information, which encourages openness and accountability in AI-driven decision-making.⁶

1.1.3 Adoption Challenges and Opportunities for SMEs

Small and medium-sized businesses (SMEs) often face a unique set of problems that stop them from getting the most out of AI technology. Some of these problems are not having enough technical knowledge, not having enough re-sources, and having trouble integrating new AI models into existing business processes. Also, small and medium-sized businesses (SMEs) often don't have the processing power, storage space, or trained staff needed to run AI systems.

Despite these challenges, SMEs can derive significant advantages from the integration of AI and XAI. By using AI technologies, small and medium-sized businesses (SMEs) can improve customer service, make their operations run more smoothly, and make better strategic decisions. XAI can make AI more accessible to small and medium-sized

⁴ Cf. Vilone, G. & Longo, L., Explainable Artificial Intelligence: A Systematic Review, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

⁵ Cf. Nemko, A., Explainable AI and Business Value Creation, *AI & Society*, Vol. 40, No. 1, 2025, pp. 89–103.

⁶ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, *Proceedings of the 22nd ACM SIGKDD Conference*, 2016, pp. 1135–1144.

businesses by giving them clear and useful information. This will help them compete with bigger companies that have access to technologies.⁷

In this case, LIME offers a useful solution by making AI-driven decisions clear without requiring SMEs to have a lot of technical knowledge. One important thing that could speed up the use of AI in small and medium-sized businesses is the ability to clearly explain AI models, especially in areas like predicting customer churn, demand forecasting, and predictive maintenance.

1.1.4 Importance of Explainability in Business Strategy

In addition to being a matter of ethics and rules, explainability in AI is important for practical business strategy. The ability of SMEs to understand and explain decisions made by AI has a direct effect on how they make strategic decisions. Small and medium-sized enterprises (SMEs) rely on decision support systems to help them make smart choices that help their businesses grow, whether they are predicting sales, improving marketing campaigns, or anticipating how customers act.⁸

LIME gives small and medium-sized businesses (SMEs) clear explanations of how AI models make predictions, which helps them understand how they work. These insights give decision-makers the power to look at model projections, figure out if they are realistic, and make the necessary changes.⁹ In industries like manufacturing and retail where speed and flexibility are important, being able to quickly understand AI-driven data is a big competitive advantage.

Companies can also better align their AI systems with their bigger strategic goals by using explainable AI methods like LIME. This alignment makes sure that AI systems are connected to business goals like improving customer satisfaction, increasing efficiency, or promoting sustainability, instead of working alone for small and medium-sized businesses. When small and medium-sized businesses (SMEs) use AI to make decisions,

⁷ Cf. Chatterjee, S. et al., Big Data Analytics and Decision Quality, Journal of Business Research, 2023.

⁸ Cf. Turban, E., Sharda, R. & Delen, D., Decision Support and Business Intelligence Systems, 10th ed., Pearson, 2014, pp. 1–25.

⁹ Cf. Power, D. J., A Brief History of Decision Support Systems, DSS Resources, 2014.

being open and responsible is becoming more important for keeping stakeholder trust and making sure the business is successful.

1.2 Problem Statement

1.2.1 Lack of Interpretability in AI Models

AI models have done quite well in many fields, especially those that use deep learning and ensemble methods. These models work quite well for things like predictive analytics, natural language processing, and recognizing pictures. Still, their "black-box" nature is a big problem.¹⁰ These models can make accurate predictions, but they don't always tell you how they got there.

For Small and Medium-Sized Enterprises (SMEs), this lack of transparency in AI decision-making can create several problems:¹¹

- **Trust Issues:** Many small and medium-sized business (SME) decision-makers don't have a lot of technical knowledge; therefore, they might not want to adopt AI technology that they don't completely comprehend. If management can't explain why the model made the judgments it did, the organization may not be able to adopt and use AI technology.¹²
- **Inability to Validate Predictions:** Small and medium-sized businesses (SMEs) often employ AI to make strategic decisions, such as projecting customer loss or demand. If the model can't explain its predictions, managers can't check or challenge the AI's suggestions. This lack of validation could lead to bad decisions, especially in important business areas.
- **Regulatory and Ethical Concerns:** If AI models can't be understood, it can lead to moral dilemmas or even legal problems in areas where AI models' decisions could have big moral, financial, or legal consequences (like approving loans,

¹⁰ Cf. Vilone, G. & Longo, L., Explainable Artificial Intelligence: A Systematic Review, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

¹¹ Cf. Mintzberg, H., *The Rise and Fall of Strategic Planning*, Free Press, 1994, pp. 3–12.

¹² Cf. Gunning, D. et al., *XAI—Explainable Artificial Intelligence*, Defense Advanced Research Projects Agency (DARPA), 2019.

making hiring decisions, or diagnosing illnesses). If SMEs can't explain how AI models make decisions, they could be at risk of things like making decisions that aren't fair or not following the rules.

To solve these difficulties, we really need XAI solutions that can make AI-driven decision-making more open. LIME is a model-agnostic method that helps SMEs understand how their models are making predictions by giving them localized explanations. This can help increase responsibility and trust.¹³

1.2.2 Inadequate Decision Support Systems in SMEs

Larger companies may have access to advanced decision support systems that work well with AI models, but small and medium-sized businesses (SMEs) generally don't have the infrastructure they need to construct or use these kinds of systems. A lot of small and medium-sized businesses (SMEs) still make decisions based on gut feelings, past experiences, and static business information tools. In a world where data-driven judgments are important for staying ahead of the competition, these methods are no longer enough.¹⁴

Several factors contribute to the inadequacy of decision support systems in SMEs:

- **Limited Resources:** SMEs generally must stick to a tight budget, which makes it hard for them to buy advanced AI-driven decision assistance solutions. These organizations might not have the technical know-how to set up and keep AI models running in their current systems.¹⁵
- **Fragmented Data:** A lot of small and medium-sized businesses still use data sources that aren't connected to a single repository. This fragmentation makes it hard to get a whole picture from the data that is accessible and makes it harder to use AI and predictive analytics models.
- **Outdated Systems:** Many small and medium-sized businesses still utilize old decision-making systems that don't work with new AI technologies. These systems

¹³ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, Proceedings of the 22nd ACM SIGKDD Conference, 2016, pp. 1135–1144.

¹⁴ Cf. Turban, E., Sharda, R. & Delen, D., Decision Support and Business Intelligence Systems, 10th ed., Pearson, 2014, pp. 1–25.

¹⁵ Cf. Villeneuve, E., Baret, S. & Courcelle, O., Multi-Source Data Integration for Decision-Making, Decision Support Systems, Vol. 165, 2023, Article 113880.

might use batch processing or static reporting, which aren't good for making decisions in real time. Because of this, small and medium-sized businesses have a hard time adapting to changes in the commercial world.

- **Complexity in AI Model Integration:** Even when SMEs see the potential of AI, adding machine learning models to their current decision support systems can be quite hard. Many small and medium-sized businesses don't have the right technical infrastructure, such cloud platforms, large data processing capabilities, and qualified workers, to use AI well.¹⁶

To get around these problems, small and medium-sized businesses (SMEs) require affordable, easy-to-use solutions that can add AI models to their decision-making processes. Adding LIME to current systems creates a link between complicated AI models and useful decision support tools. This makes it easy for small and medium-sized businesses (SMEs) to make smart, data-driven decisions.

1.2.3 Lack of Trust in AI-Driven Decisions

Trust is an important part of how every organization makes decisions. AI systems need people to trust the model's predictions and suggestions in order to work. People lose faith in AI because they don't understand how it makes decisions. This is especially bad for small and medium-sized businesses (SMEs) because their managers may not have the technical knowledge to understand how AI models work.¹⁷

The lack of trust in AI-driven decisions is particularly problematic in SME contexts for several reasons:

- **Fear of Bias:** AI models, especially those based on historical data, can inadvertently perpetuate biases present in the training data. If SMEs cannot understand the factors influencing AI decisions, they may fear that the models are producing biased or unfair predictions.

¹⁶ Cf. Schwaeppe, J., *The New Normal: The Status Quo of AI Adoption in SMEs*, Technological Forecasting and Social Change, Vol. 198, 2025.

¹⁷ Cf. Peretz-Andersson, M., Berglund, F. & Johansson, B., *Artificial Intelligence Implementation Challenges in Manufacturing SMEs*, Journal of Manufacturing Technology Management, Vol. 35, No. 1, 2024, pp. 112–130.

- **Resistance to Change:** Many small and medium-sized businesses (SMEs) don't want to adopt AI systems since they are used to making judgments in more traditional methods. Workers and managers won't want to use AI systems until they can understand why the models make the predictions they do.
- **Accountability Issues:** In business contexts where AI-driven decisions could have big moral or financial effects, it's very important to make sure that people are held accountable. Unless there are clear causes, it's impossible to blame the AI system or its creators for mistakes or faulty predictions. People are hesitant to use AI for key decisions because they are worried that it won't be responsible.

LIME helps to solve these trust issues by giving clear, local explanations for certain forecasts. Small and medium-sized businesses (SMEs) can feel more confident in AI systems if they know why forecasts were made and what factors affected them.¹⁸

1.2.4 Ethical and Regulatory Challenges

AI judgments have a tremendous impact on areas like finance, healthcare, and human resources. Because of this, the social, legal, and moral repercussions of employing AI are very important. A lot of small enterprises don't recognize that AI systems can cause ethical difficulties. Also, it can be hard for firms to meet norms that call for openness, justice, and responsibility because AI models are sometimes hard to explain.

For example, AI is being utilized more in the financial services industry to monitor risk, approve loans, and check credit scores. Not being able to explain how an AI model makes its choice in certain cases can lead to accusations of discrimination or unjust treatment. People are also worried about fairness and bias when recruiting people because AI is being used in hiring processes. If SMEs don't make it clear how AI models pick candidates, they could be sued for discrimination and other legal problems.

LIME's ability to explain AI assertions clearly is highly helpful in these instances. LIME helps small and medium-sized enterprises stay out of trouble by making it obvious how

¹⁸ Cf. Nemko, A., Explainable AI and Business Value Creation, *AI & Society*, Vol. 40, No. 1, 2025, pp. 89–103.

AI models make decisions. It helps businesses make sure that their AI systems are moral and follow the law. This makes customers, employees, and regulators trust you more.

1.3 Research Objectives

1.3.1 Objective 1: Investigating the Application of LIME in SME Decision-Making

This objective aims to explore how LIME can be applied within SMEs to improve decision-making processes. Specifically, the study will:

- Assess how LIME enhances model interpretability, providing localized explanations for AI predictions in areas like customer churn, resource allocation, and sales forecasting.
- Evaluate the ease of use of LIME for non-technical SME decision-makers, ensuring that LIME-generated explanations are accessible and actionable for users without deep technical expertise.

1.3.2 Objective 2: Assessing the Impact of LIME on Trust and Transparency

This objective focuses on determining how LIME impacts the transparency and trustworthiness of AI predictions in SMEs. Key areas of focus will include:

- Evaluating how LIME helps build trust in AI-driven decisions by providing clear, understandable explanations.
- Assessing whether LIME increases confidence in decision-making by offering transparency and clarity in AI model predictions.

1.3.3 Objective 3: Comparing LIME with Other XAI Techniques

This objective aims to compare LIME with other widely used explainable AI techniques, such as SHAP and Partial Dependence Plots (PDPs). The comparison will assess:

- Interpretability: How well each method explains AI predictions for SMEs.
- Computational efficiency: The feasibility of each technique for SMEs with limited resources.

1.3.4 Objective 4: Developing a Framework for LIME Integration

The final objective is to propose a **framework** for integrating LIME into existing decision support systems in SMEs. This framework will ensure that LIME can be implemented in a cost-effective, user-friendly manner that enhances decision-making without requiring extensive technical expertise.

1.4 Research Questions

1.4.1 Primary Research Question

How can the integration of Local Interpretable Model-agnostic Explanations (LIME) enhance decision-making in Small and Medium-Sized Enterprises (SMEs)?

This question aims to explore the core focus of the thesis: the application of LIME to improve transparency and trust in AI-driven decisions within SMEs, and how it can aid decision-making processes.

1.4.2 Sub-Questions

1. How does LIME improve the interpretability and transparency of AI predictions for SME decision-makers?

This sub-question examines the effectiveness of LIME in making AI predictions understandable, and how it helps SME managers interpret and trust model outputs.

2. What impact does LIME have on the trust and confidence of SME managers in AI-driven decision-making?

This will assess whether LIME's explanations increase the confidence of decision-makers in the use of AI models for business strategy and operational decisions.

3. How does LIME compare with other explainable AI techniques, such as SHAP, in terms of interpretability and computational efficiency?

This question compares LIME with alternative methods to determine which is most suitable for SMEs, focusing on ease of use and resource requirements.

4. What framework can be developed for the successful integration of LIME into SME decision support systems?

This sub-question focuses on creating a practical framework for integrating LIME into SME operations, ensuring that it is technically feasible and user-friendly.

1.5 Justification for the Study

This study is important for several reasons, particularly in the context of AI adoption within SMEs, and the growing need for transparent, understandable, and accountable AI-driven decision-making.

1.5.1 Bridging the Gap Between AI and SMEs

AI has a lot of potential to help small and medium-sized businesses (SMEs) work better, save money, and make better strategic decisions. But because AI models are so complicated, many small and medium sized businesses are still hesitant to use AI powered systems. Decision-makers cannot effectively leverage AI technologies because machine learning models are often regarded with scepticism due to their complexity and ambiguity.¹⁹

This study aims to reconcile the practical requirements of SMEs with the intricate functionalities of AI models by concentrating on Local Interpretable Model-agnostic Explanations (LIME). LIME gives short, clear explanations for each forecast. This makes it easier for SME managers, who may not be very tech-savvy, to understand, trust, and use AI-driven insights. This study will show how small and medium-sized businesses (SMEs) can get the most out of artificial intelligence (AI) by using it in a clear and easy-to-understand way.²⁰

¹⁹ Cf. Schwaeppe, J., *The New Normal: The Status Quo of AI Adoption in SMEs, Technological Forecasting and Social Change*, Vol. 198, 2025.

²⁰ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, *Proceedings of the 22nd ACM SIGKDD Conference*, 2016, pp. 1135–1144.

1.5.2 Addressing the Lack of Trust in AI Decision-Making

Trust is an important part of every decision-making process. Small and medium-sized businesses (SMEs) may see AI as a "black box," which makes them unsure about how accurate and fair AI predictions are. Decision-makers are reluctant to adopt AI systems for critical business decisions, such as customer retention, resource allocation, and financial forecasting, due to a lack of clear explanations.²¹

This study will contribute to the literature on Explainable AI (XAI) by illustrating how LIME could enhance trust in AI-driven decision-making. LIME boosts decision-makers' trust in AI by giving clear and understandable reasons for predictions. This makes them more likely to use AI in strategic processes. The study aims to illustrate that transparency, enabled by techniques such as LIME, is crucial for bolstering trust in AI models.²²

1.5.3 Contributing to the Field of XAI

The future of using AI in business depends on the growing field of Explainable Artificial Intelligence (XAI). As more businesses use AI in their daily operations, the need for transparency and interpretability grows. This is especially true for businesses where AI decisions have big ethical, financial, and legal effects. This study enhances the field of XAI by focusing on a practical application, LIME, that aids non-technical decision-makers in SMEs in understanding AI more effectively.²³

This study also adds to the body of current work by comparing LIME to other XAI methods (like SHAP) in the context of SMEs. The findings will delineate a framework for subsequent research and applications of XAI in SMEs by clarifying the most effective methodologies for diverse organizational contexts and decision-making processes.

²¹ Cf. Vilone, G. & Longo, L., Explainable Artificial Intelligence: A Systematic Review, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

²² Cf. Nemko, A., Explainable AI and Business Value Creation, *AI & Society*, Vol. 40, No. 1, 2025, pp. 89–103.

²³ Cf. Gunning, D. et al., XAI—Explainable Artificial Intelligence, Defense Advanced Research Projects Agency (DARPA), 2019.

1.5.4 Practical Implications for SMEs

The findings of this study will have practical implications for SMEs that are considering the adoption of AI technologies, but face challenges related to model complexity and trust. By demonstrating how LIME can make AI-driven insights more accessible, transparent, and actionable, this research will provide SMEs with a blueprint for incorporating AI into their decision-making processes in a way that is both practical and trustworthy.

The research will also highlight how SMEs can overcome common barriers to AI adoption, such as limited technical resources and expertise, by using lightweight and interpretable AI methods like LIME.

1.6 Structure Of the Thesis

This thesis is structured into six main chapters, each addressing a key aspect of the research. The following is an overview of the chapters:

- **Chapter 1: Introduction**

This chapter introduces the background of the study, outlining the context, problem statement, research objectives, research questions, and the justification for the research. It also describes the scope of the study and provides an overview of the thesis structure

- **Chapter 2: Literature Review**

This chapter reviews existing literature on Explainable AI (XAI), focusing on techniques such as LIME and SHAP. It explores the challenges SMEs face in adopting AI, the need for transparency in AI-driven decision-making, and the role of XAI in overcoming these challenges. The review also identifies gaps in the literature and positions this research within the broader context of AI in SMEs.

- **Chapter 3: Methodology**

This chapter outlines the research design and methodology used in the study. It discusses the approach to data collection, including the selection of case studies and SMEs, as well as the methods for implementing and evaluating LIME in

decision support systems. The chapter also describes the tools and techniques used for data analysis.

- **Chapter 4: Results**

In this chapter, the findings from the application of LIME in SME decision-making are presented. The chapter focuses on how LIME impacts the transparency and interpretability of AI models, and how it influences decision-making in practice. Results are presented using both qualitative and quantitative data, illustrating the practical effectiveness of LIME.

- **Chapter 5: Discussion**

This chapter analyses the results in the context of the research objectives and questions. It compares LIME with other XAI techniques and discusses the implications for SMEs. The chapter also explores the challenges faced in integrating LIME and provides insights into the broader applicability of XAI for SMEs in different industries.

- **Chapter 6: Conclusion**

The final chapter summarizes the key findings of the research, addresses the research questions, and discusses the contributions to the field of XAI and SME decision-making. It also offers recommendations for SMEs looking to adopt AI and provides suggestions for future research.

2. Literature Review

2.1 SMEs, Big Data and AI in Strategic Decision-Making

Small and medium-sized firms (SMEs) are an important aspect of national and regional economies, especially in Europe, where they make up most of the business activity, job creation, and potential for innovation. Their strategic importance is becoming clearer as digital transformation continues, and the ability to use data-driven insights is becoming a key factor in long-term competitiveness. Big Data Analytics (BDA) and Artificial Intelligence (AI) have become more important for small and medium-sized businesses (SMEs) in the last few years. However, not all SMEs have adopted them yet. This section examines current academic perspectives on the role of AI and big data in SME decision-making, the determinants influencing adoption, the strategic outcomes linked to BDA utilization, and the enduring limitations that affect SMEs' interaction with AI-enabled decision support.²⁴

To contextualize the relevance of explainable, data-driven decision support within small and medium-sized enterprises, the key organizational characteristics of SMEs and their implications for strategic decision-making are summarized in Table 1.

TABLE 1: Characteristics of SMEs and Implications for Strategic Decision-Making

SME Characteristic	Description	Implication for Strategic Decision-Making
Limited analytical resources	SMEs typically lack dedicated data science teams and advanced IT infrastructure	Decision-support systems must be simple, cost-effective and interpretable
Centralized managerial authority	Strategic decisions are often made by owners or a small leadership group	AI outputs must support individual managerial

²⁴ Cf. Štrukelj, T., Šuligoj, M., Lesjak, D. & Podbregar, I., Small- and Medium-Sized Enterprises: The Role of Artificial Intelligence in Decision Making, *Administrative Sciences*, Vol. 15, No. 7, 2025, pp. 1–21.

		judgement rather than automated decision-making
Informal decision processes	Decisions are frequently based on experience and intuition rather than formal analytics	Explainable AI is necessary to align model outputs with managerial reasoning
Limited tolerance for risk	Strategic errors can have immediate financial consequences	Transparent and interpretable models increase trust and adoption

Source: Own Synthesis

An increasing amount of study investigates how AI and BDA help SMEs make decisions at different levels of the organization. They characterize AI as a transformative tool that can augment both operational and strategic decisions, contending that the technology presents novel options for informed judgment, enhanced efficiency, and competitive positioning. Their investigation underscores that AI is progressively utilized to enhance functions such as forecasting, demand planning, client segmentation, and inventory optimization, areas that were once dependent on managerial intuition or constrained historical analysis. A paper similarly shows that BDA has a substantial impact on the quality of decisions. They show that companies that use big data capabilities have better business performance and more accurate forecasts. Their empirical findings substantiate the assertion that BDA is not merely a technological instrument but a facilitator of enhanced managerial judgment, particularly in contexts marked by uncertainty.²⁵

This connection between data-driven insights and strategic effectiveness is backed up by research that looks at bigger organizational results. In a comprehensive study of the effects of big data on the performance of small and medium-sized enterprises (SMEs), finds that using analytics makes companies more flexible, sensitive to market changes, and able

²⁵ Cf. Chatterjee, S., Rana, N. P., Tamilmani, K. & Sharma, A., Big Data Analytics and Decision-Making Quality: Evidence from Emerging Firms, *Information Systems Frontiers*, Vol. 25, 2023, pp. 1–17.

to make better use of their resources.²⁶ The review delineates many ways by which big data affects performance, such as augmented consumer insight, refined process optimization, and increased capacity for anticipatory decision-making. These findings are consistent with the broader academic perspective that analytics enhances firm-level dynamic capacities, enabling organizations to perceive, seize, and reorganize resources based on data-driven insights. From this point of view, AI and BDA are important parts of strategic management since they help SMEs find new possibilities, predict dangers, and improve their strategic direction.

Even though these benefits seem clear, small and medium-sized businesses (SMEs) still don't use AI consistently. The level of implementation maturity varies a lot by sector and area paper investigates the current state of AI adoption in SMEs, revealing that, despite a high awareness of AI's potential, real implementation is limited by structural and organizational challenges.²⁷ A lot of small and medium-sized businesses (SMEs) only have basic data infrastructures and not very good analytical skills, which makes it harder for them to use advanced AI systems. The survey identifies cost, a shortage of skilled staff, insufficient digital readiness, and uncertainty regarding AI's practical effects as primary obstacles to adoption. This aligns with Mathagu's findings, which stress that effective AI implementation in SMEs relies on a blend of technology preparedness, organizational dedication, and external environmental influences.²⁸ These studies together indicate that the adoption of AI in SMEs is influenced not just by technological availability but also by internal capabilities, cultural attitudes, and resource limitations.

To formalize the conceptualization of these adoption drivers, numerous scientists utilize established theoretical models, notably the Technology–Organization–Environment (TOE) framework. A paper shows, utilizing the Technology-Organization-Environment (TOE) framework in the context of Big Data Analytics (BDA) adoption in small and medium-sized enterprises (SMEs), contend that technological attributes, including

²⁶ Cf. Kgakatsi, I. B., The Impact of Big Data Analytics on SME Performance: A Systematic Literature Review, *Journal of Small Business and Enterprise Development*, Vol. 31, No. 2, 2024, pp. 245–268.

²⁷ Cf. Schwaeke, J., The New Normal: The Status Quo of Artificial Intelligence Adoption in SMEs, *Technological Forecasting and Social Change*, Vol. 198, 2025.

²⁸ Cf. Mathagu, T., Determinants of Artificial Intelligence Adoption in Small and Medium Enterprises, *Journal of Business Research*, Vol. 167, 2024, pp. 113–125.

compatibility, complexity, and perceived utility, impact the preliminary decision to embrace analytics tools.²⁹ Organizational factors, including executive backing, financial resources, and human capital, determine the degree to which adoption may be implemented. Finally, outside variables like rivalry in the business, government rules, and customer expectations can either help or hurt adoption. The TOE model's layered structure is great for looking at the many different problems that small and medium-sized businesses (SMEs) confront. It also helps us understand why adoption is still inconsistent even though AI's potential usefulness is widely recognized.

In addition to adoption drivers, research also shows the distinct strategic advantages that AI offers to small and medium-sized businesses (SMEs). A common theme is improving the speed and quality of decisions. The Wiley report on Big Data Analytics Adoption and SME Market Performance (2025) says that SMEs who use BDA do better in the market, especially when it comes to reaching more customers, making more money, and standing out from the competition.³⁰ Their research shows that analytics capabilities enable businesses react more quickly to changes in the market and customer needs. This suggests that BDA is a strategic asset. This aligns with previous research who advocates for a strategic BDA framework that regards analytics as a pivotal resource for opportunity recognition and strategic planning.³¹ Their methodology stresses the need of using descriptive, predictive, and prescriptive analytics in strategic processes. They say that small and medium-sized businesses (SMEs) may gain a lot from making analytics a regular part of how they do business, rather than just something they do occasionally.

AI's effect is most clear in judgments about money and resources. The 2024 study "Artificial Intelligence in SME Financial Decision-Making" shows that AI technologies are being used more for things like credit rating, profitability analysis, budgeting, and risk assessment. These applications enable finance managers at SMEs to assess scenarios more methodically and to detect potential concerns sooner. The research emphasizes the

²⁹ Cf. Maroufkhani, P., Wagner, R., Wan Ismail, W. K. & Nourani, M., Big Data Analytics Adoption Model for SMEs: A TOE Perspective, *Journal of Small Business Management*, Vol. 58, No. 4, 2020, pp. 1–30.

³⁰ Cf. Wiley Analytics Research Team, *Big Data Analytics Adoption and SME Market Performance*, Wiley Online Library, 2025.

³¹ Cf. Willetts, R., Sullivan, R. & James, P., A Strategic Framework for Big Data Analytics in SMEs, *Strategic Change*, Vol. 29, No. 6, 2020, pp. 613–628.

significance of transparency and trust, indicating that SME decision-makers are more inclined to utilize AI tools when model outputs are comprehensible and align with domain expertise. This finding is particularly pertinent to the current thesis, which underscores explainability as a fundamental element for effective decision support.³²

Nonetheless, some experts contend that the advantages of AI for SMEs are still aspirational rather than completely actualized. Concentrating on the adoption of AI in manufacturing SMEs, discovering that actual deployments frequently encounter difficulties stemming from fragmented data sources, inadequate organizational coordination, and constrained change management capabilities. Their research indicates that SMEs might effectively use standalone AI solutions yet struggle to incorporate them into unified strategic workflows. This fragmentation leads to suboptimal performance compared to AI's complete capabilities. The authors advocate for comprehensive frameworks that amalgamate technical capabilities with organizational processes and human skills, emphasizing the necessity for methodologies that can connect predictive analytics with actionable managerial insights.

A persistent concern in the literature is the deficiency of transparency and interpretability in AI-driven decision support systems utilized by SMEs. Although numerous studies illustrate that AI and BDA can improve decision-making, few research examines how SME managers, who generally lack specialized analytics expertise, interpret the outputs of intricate models. Decision-makers frequently lack clarity on the generation of AI-generated recommendations, thus diminishing trust and hindering adoption.³³ The Wiley (2025) report also stresses that for SMEs to get the most out of analytics, the insights must be easy for people who aren't tech-savvy to understand. This is in line with other worries in the area that "black-box" or unclear models could make things riskier by making decisions that those who have to carry them out don't understand.³⁴

³² Cf. Artificial Intelligence in SME Financial Decision-Making, Emerging Applications and Challenges, *Journal of Financial Innovation*, Vol. 10, No. 1, 2024, pp. 45–61.

³³ Cf. Štrukelj, T. et al., Small- and Medium-Sized Enterprises: The Role of Artificial Intelligence in Decision Making, *Administrative Sciences*, Vol. 15, No. 7, 2025.

³⁴ Cf. Wiley Analytics Research Team, *Big Data Analytics Adoption and SME Market Performance*, Wiley Online Library, 2025.

The literature collectively indicates that AI and big data provide significant potential for SMEs; yet obstacles to adoption, constrained organizational capabilities, and insufficient explainability hinder their efficient utilization. Although current research has examined adoption models, performance outcomes, and strategic implications, there is a scarcity of integrated frameworks that amalgamate multi-source data analytics, predictive modelling, and explainable outputs in a manner that corresponds with SME capabilities and decision-making processes. This discovery necessitates the development of new models, such as the framework proposed in this thesis, that clearly integrate explainability as a fundamental necessity, rather than as an optional or supplementary element, inside SME decision support settings.

2.2 Big Data, Multi-Source Data and Decision Support Systems

The rise of digital technology and the spread of processes that create data inside and outside of businesses have changed the way companies make decisions in a big way. Big data has changed the way people make decisions from relying on gut feelings to relying on facts and analysis. Increasingly, scholarly work is stressing that the strategic value of big data isn't just having access to information, but also being able to combine, understand, and use different data sources in decision support contexts. This part talks about how big data has changed the way decisions are made, the importance of combining data from many sources, and the research on decision support systems (DSS) and business intelligence (BI) as tools for helping organizations learn and plan their strategies.

People in many fields agree that big data plays a big role in making decisions in organizations. Massaro (2023) talks about how big data has become a key part of modern decision-making frameworks that help businesses turn raw data into useful information.³⁵ Their evaluation asserts that big data is crucial for improving the timeliness, precision, and depth of decisions, contending that organizations proficient in utilizing analytics acquire substantial strategic benefits. These benefits include better forecasting, a better understanding of customers, more efficient operations, and the ability to spot new trends

³⁵ Cf. Massaro, M., Big Data and Decision-Making: A Structured Literature Review, *Management Decision*, Vol. 61, No. 3, 2023, pp. 673–695.

before they happen. An author contends that big data facilitates organizations in making decisions that are more contextually informed by utilizing a variety of information sources, such as transactional data, market signals, environmental indicators, and behavioural patterns.³⁶ Their findings indicate that the amalgamation of local, multi-source data can substantially enhance decision quality by offering a more comprehensive depiction of organizational circumstances.

A fundamental difficulty recognized in the literature is the amalgamation of data from many sources. Increasingly, businesses depend on data from their own systems (including ERP, CRM, and HRIS), outside platforms (like social media and competitor databases), IoT devices, consumer interactions, and market intelligence services. In a thorough evaluation of data integration strategies, an author says that multi-source integration has become a basic need for good decision support.³⁷ The report says that the main problems include dealing with data that moves quickly, comes in many forms, is inconsistent, and is missing information. Balaha says that the quality of decision-making is strongly related to how strong data integration architectures are, especially when companies rely on analytical models that need huge, diversified datasets. Their work underscores the increasing significance of contemporary data architectures, such as data lakes, lakehouses, and federated data models, that can adaptively integrate structured and unstructured data.

The literature also stresses a lot the connection between the quality of information and the results of decisions. Hjelle et al., who looked at how organizations make decisions, say that the quality of the information they use, such as how accurate, complete, timely, and easy it is to get, is a key factor in the quality of the decisions they make.³⁸ Their real-world evidence shows that even advanced analytics technologies don't help make better decisions when the data they use is broken up or not consistently connected throughout departments. This conclusion supports the idea that integrating data from several sources

³⁶ Cf. Villeneuve, E., Baret, S. & Courcelle, O., Leveraging Local and Multi-Source Data for Strategic Decision-Making, *Decision Support Systems*, Vol. 165, 2023, Article 113880.

³⁷ Cf. Balaha, M., A Review of Multi-Source Data Integration Techniques for Decision Support, *Journal of Big Data*, Vol. 12, No. 1, 2025.

³⁸ Cf. Hjelle, K., Røstad, L. & Waaler, A., Information Quality as a Determinant of Decision-Making Performance, *Information Systems Management*, Vol. 41, No. 1, 2024, pp. 52–66.

is not only a technical task, but a strategic skill that affects how organizations learn and compete.

Decision support systems (DSS) have been used for a long time as both theoretical and practical tools to help us understand how data, models, and users work together in decision-making. A paper characterizes DSS as systems engineered to facilitate managerial decision-making through the provision of analytical tools, models, and organized information.³⁹ They say that current DSS use a mix of descriptive, predictive, and prescriptive analytics, which makes them useful for a wide range of strategic and operational purposes. The DSS literature stresses how important it is to make sure that analytical tools meet the demands of users. It also stresses that decision-makers should be able to look at data, test their assumptions, and look at other possible outcomes. In this way, DSS are the ideas that make up the foundation of data-driven decision-making, connecting the technical parts of analytics to the way managers think and make decisions.

The transition of DSS into AI-enabled systems is a notable advancement in the literature. Soori and group says that AI-based DSS, especially in Industry 4.0 settings, make it easier for decision-makers to sort through big datasets, find trends, and get automated suggestions.⁴⁰ Their review shows that AI-DSS systems can provide insights that traditional analytics methods can't, especially when relationships are complicated and not linear. The authors also admit that the complexity of AI creates additional problems, such as model opacity, over-reliance on automated advice, and the possibility that human expertise would be pushed to the side. This makes it clear how important it is to include explainability features in DSS, which is a topic that will be covered in greater detail in Section 2.3.

Along with conceptual frameworks, empirical studies show the pros and cons of using DSS and multi-source analytics in real life. The exploratory study from 2020 on the effects of BI and DSS demonstrates that companies who use BI tools make better decisions

³⁹ Cf. Turban, E., Sharda, R. & Delen, D., *Decision Support and Business Intelligence Systems*, 10th ed., Pearson, 2014, pp. 35–62.

⁴⁰ Cf. Soori, P. K., Arezoo, B. & Dastres, R., *Artificial Intelligence-Based Decision Support Systems in Industry 4.0: A Review*, *Computers & Industrial Engineering*, Vol. 183, 2024, Article 109466.

because they have better access to information, better visualization, and better analytical help. The authors say that organizations that use BI are better able to find performance gaps, keep an eye on KPIs, and come up with strategic responses. But the study also points out problems that come up when data is kept in separate places, which stops companies from fully using BI's potential. These results fit with what has been found in other studies, which show that just having technology tools isn't enough without data integration and quality assurance.⁴¹

As analytics systems have gotten increasingly complicated, multi-source data fusion has been a topic of great interest to researchers. In a study on multi-source data fusion for trustworthy decision-making, they provide an optimized architecture that combines several datasets for better analytical results.⁴² Their work is mostly on IoT and sensor-based environments, but the ideas they present can be used in many other situations. They show how fusion approaches can combine conflicting or redundant information from many sources to make sure that analytics get dependable, high-quality data inputs. Their results show that multi-source fusion not only makes algorithms work better, but it also makes decision-makers surer that the analytical outputs are correct. New benchmarks like FDABench (2025) back up these ideas by showing the technical problems that current data agents face when they have to manage complicated analytical workflows that involve a lot of different sorts of data. FDABench emphasizes the increasing significance of automation, orchestration, and consistency in multi-source analytical contexts, while it is not specific to SMEs.⁴³

A study stresses the importance of multi-source data in making strategic decisions. They say that decisions based on local, contextual data are usually more accurate and useful.⁴⁴ Their research suggests methodology wherein multi-source analytics facilitate the integration of internal performance indicators with external environmental signals within

⁴¹ Cf. Popovič, A., Hackney, R., Coelho, P. S. & Jaklič, J., How Information Quality Mediates the Impact of Business Intelligence on Decision-Making, *Decision Support Systems*, Vol. 62, 2020, pp. 19–32.

⁴² Cf. Khan, S., Yao, X. & Wang, Z., Multi-Source Data Fusion for Trusted Decision-Making, *IEEE Access*, Vol. 13, 2025, pp. 112340–112355.

⁴³ Cf. Massaro, M., Big Data and Decision-Making: A Structured Literature Review, *Management Decision*, Vol. 61, No. 3, 2023.

⁴⁴ Cf. Villeneuve, E. et al., Leveraging Local and Multi-Source Data for Strategic Decision-Making, *Decision Support Systems*, 2023.

organizations. This kind of integration is especially useful for small and medium-sized businesses (SMEs) that work in fast-changing and competitive markets where being aware of what's going on can be very important. But, as the authors point out, many small and medium-sized businesses (SMEs) don't have the skills they need to apply it well, such as data engineering knowledge, advanced analytics skills, and tools for processing unstructured or semi-structured data.

Even while multi-source big data is clearly useful, the research shows that it is still not being used enough in small businesses. Numerous studies indicate that SMEs face challenges in adopting sophisticated analytics due to financial limitations, insufficient technical knowledge, and a lack of understanding with data engineering prerequisites. Small and medium-sized businesses (SMEs) are very interested in using big data tools, but their ability to do so is limited by their organizational capacity and digital maturity. As a result, small and medium-sized businesses (SMEs) commonly use analytics technologies in incomplete or shallow ways, which limits their strategic influence.

A common idea in the literature is that multi-source analytics by itself does not ensure better decision-making. Instead, companies need systems that can turn integrated data into useful insights and show them in ways that make sense to managers. This finding highlights how important DSS and BI tools are as middlemen between complicated data infrastructures and user-centered decision-making. Dashboards and BI front ends are very important for making data from many sources easy for managers to find and comprehend.

In short, the research shows that big data and multi-source analytics can change the way businesses make decisions, but to take advantage of these opportunities, businesses need to have strong data integration, high-quality information, and tools that turn analytics results into useful insights for managers. DSS and BI frameworks offer robust conceptual underpinnings; yet, current research indicates deficiencies in the integration of advanced AI models, especially within SME contexts. AI algorithms are hard to understand, and the fact that data is spread out makes it even harder. This makes it hard for managers to use the data they have. This shows how important explainability and user-centered design

are for small and medium-sized businesses (SMEs) to make good decisions based on data. We will go into more detail about these subjects in the next section.⁴⁵

2.3 Explainable AI in Business Analytics and Decision-Making

As AI becomes more common in the decision-making processes of businesses, worries about trust, transparency, and interpretability have grown in both academic and managerial settings. Machine learning models can make predictions and analyse data much more accurately, but they are typically not very useful in business settings since they are hard to understand. Many common algorithms, such as ensemble approaches and non-linear predictive models, work like "black boxes," giving results without explaining how they got there in a way that makes sense. In situations where decisions need to be made based on responsibility, rationale, and managerial judgment, this lack of transparency can hurt trust, slow down adoption, and make AI-driven insights less effective in real life. Explainable Artificial Intelligence (XAI) has thus become an essential response to the increasing dependence on intricate models, seeking to render algorithmic choices comprehensible, evaluable, and actionable for human decision-makers.

In business analytics, where AI systems are increasingly used to help make strategic, tactical, and operational choices, the requirement for explainability is especially strong. Business decision-making is different from purely technical contexts because it contains many people, conflicting goals, and contextual factors that model accuracy alone can't fully reflect. A study contends that in the absence of explainability, AI systems may become impractical in critical areas, as decision-makers are unable to assess whether model suggestions are consistent with organizational objectives, ethical norms, or domain knowledge.⁴⁶ This problem is much worse in small and medium-sized businesses (SMEs), where a small group of managers often has the power to make decisions but may not have the specialized analytics knowledge needed to do so. In these situations, AI outputs must

⁴⁵ Cf. Villeneuve, E. et al., *Leveraging Local and Multi-Source Data for Strategic Decision-Making*, Decision Support Systems, 2023.

⁴⁶ Cf. Gunning, D. et al., *XAI—Explainable Artificial Intelligence*, Defense Advanced Research Projects Agency (DARPA), 2019.

be understandable not just to data scientists but also to business users, who are the ones who will have to make judgments and defend them.

Explainable AI is a set of approaches and design principles that help people understand how and why AI models give certain results. The idea of explainability, on the other hand, is more complex than just being open. A group point out that explanations can be different in many ways, such as their breadth and how they depend on the structure of the model.⁴⁷ One of the most frequently accepted differences is between global and local explanations. Global explanations try to explain how a model works across a whole dataset. They give us information about general patterns and how features are related to each other. Local explanations, on the other hand, try to explain each prediction by showing why a certain outcome happened for a certain case. In business analytics, particularly within SME contexts, local explanations tend to be more operationally pertinent, as decisions typically involve specific clients, transactions, or scenarios rather than abstract model behaviour.

Another key difference in XAI is between approaches that work with specific models and those that work with any model. Model-specific approaches depend on knowing how a certain algorithm works inside and out, while model-agnostic methods can be used with any model architecture and can be used with any prediction technique. In applied business settings, the literature is growing in favour of model-agnostic approaches since they are more flexible and less reliant on specialized technical knowledge. This adaptability is especially useful for small and medium-sized businesses (SMEs) because they may try out multiple prediction models over time but don't have the money to redesign explainability methods for each new algorithm.

From an organizational standpoint, explainability is progressively characterized as a human-centric necessity rather than only a technical attribute. Longo et al. (2024) contend that explanations ought to be assessed not solely on their fidelity to model behaviour, but also on their efficacy in enhancing human comprehension, trust, and decision-making

⁴⁷ Cf. Vilone, G. & Longo, L., *Explainable Artificial Intelligence: A Systematic Review*, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

quality.⁴⁸ This change shows that more people are realizing that explanations need to fit with how managers think, how they reason about their field, and how they make decisions. In business settings, an explanation that is mathematically accurate but hard to understand may not help people make better decisions. On the other hand, explanations that are clear and relevant to the situation, even if they are simplified, can be very useful since they help decision-makers think critically about AI results.

In this changing world, local explanation methods have gotten a lot of attention since they help with case-specific reasoning. A lot of business uses of AI involve choices that are fundamentally local, including figuring out why a certain customer is likely to leave, why a transaction is considered high risk, or why a certain forecast value is particularly high or low. In these cases, people who make decisions need explanations that connect forecasts to events that led to them. This lets them judge how likely they are, plan interventions, and explain their actions. Local explanations serve as a conduit between algorithmic inference and managerial judgment, converting model outputs into practical insights.

Local Interpretable Model-agnostic Explanations (LIME), which was first proposed by Ribeiro et al. (2016), is one of the most well-known local, model-agnostic explainability strategies. LIME is founded on the idea that a predictive model can be quite complicated on a global scale, but its behaviour can frequently be approximated locally around a given instance using a simpler, more understandable model. The method works by making disturbed samples close to the instance that has to be explained and fitting a surrogate model, usually linear, that mimics the complicated model's behaviours in that area. The feature weights show which variables had the biggest effect on the prediction. This gives a clear explanation that can be understood by people who aren't technical.⁴⁹

LIME is popular in business analytics because it is easy to use, adaptable, and focused on the local area. LIME can be used with many different algorithms because it doesn't

⁴⁸ Cf. Longo, L., Goebel, R., Lecue, F. & Kieseberg, P., *Explainable Artificial Intelligence (XAI) 2.0: Foundations, Challenges and Future Directions*, Information Fusion, Vol. 94, 2024, pp. 1–20.

⁴⁹ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, *Proceedings of the 22nd ACM SIGKDD Conference*, 2016, pp. 1135–1144.

depend on any specific model structure.⁵⁰ This makes it a good choice for small and medium-sized businesses (SMEs) that may care more about predicting performance than they do about being able to create custom explainability solutions. Also, LIME's focus on instance-level explanations is quite similar to how managers make decisions, which typically involves looking at situations instead of general trends. LIME helps people make sense of AI outputs by showing feature contributions in a way that people can understand. This lets decision-makers question AI outputs instead of just accepting them.⁵¹

LIME has shown that it works for both classification and regression tasks, which are common in business analytics. In classification scenarios, like predicting churn, categorizing risk, or rating leads, LIME can show which features push a certain instance into a certain class. This makes it possible to make more targeted decisions and interventions. LIME helps explain why a predicted value is high or low in regression contexts like sales forecasting, cost estimate, or demand prediction by finding the variables that affect it. This dual applicability makes LIME more useful in small and medium-sized enterprises (SMEs), where it is typically better to have one explainability tool that can handle various analytical jobs than to have to keep separate tools for each task.

The literature acknowledges certain limits of LIME, despite its virtues. One worry is stability: LIME uses random changes; thus, explanations may be different from one run to the next, especially in feature spaces that are very high-dimensional or very non-linear. This variety can be a problem for those who must make decisions because they want explanations to be the same every time, especially when explanations are used for audits or to justify decisions.⁵² LIME only gives a local estimate of how a model works, which is another restriction. This is usually enough for making practical decisions, but it could lead to confusion if users think that local explanations are the same as global model reasoning. Scholars like Longo et al. (2024) warn that explainability approaches need to be used with the right instructions to avoid being too sure of yourself or using them wrong.⁵³

⁵⁰ Cf. Ribeiro, M. T. et al., “Why Should I Trust You?”, 2016.

⁵¹ Cf. Lundberg, S. M. & Lee, S.-I., A Unified Approach to Interpreting Model Predictions, 2017.

⁵² Cf. Nemko, A., Explainable AI and Business Value Creation, *AI & Society*, Vol. 40, No. 1, 2025, pp. 89–103.

⁵³ Cf. Longo, L. et al., Explainable Artificial Intelligence (XAI) 2.0, *Information Fusion*, 2024.

Given the growing importance of interpretability in AI-driven decision support, a conceptual comparison of leading explainable AI techniques commonly discussed in the literature is presented in Table 2, highlighting their relative suitability for SME decision-making contexts.

TABLE 2: Comparison of Explainable AI Techniques for SME Decision

Support

Criteria	LIME	SHAP
Explanation scope	Local, instance-level explanations	Local and global explanations
Model dependency	Fully model-agnostic	Model-agnostics with model-specific optimizations
Interpretability for non-technical users	High, intuitive feature contributions	Medium, requires statistical familiarity
Suitability for SME decision-making	High, supports case-specific reasoning	Moderate, stronger for analytical teams
Practical implementation effort	Moderate	High

Source: Own synthesis based on Ribeiro et al. (2016), Lundberg and Lee (2017).^{54, 55}

Another problem is the balance between integrity and interpretability. LIME purposefully makes model behaviours simpler so that explanations are easier to grasp, however this simplification could leave out complicated interactions that are included in the model itself. In corporate settings, this trade-off is not inherently problematic, if explanations are presented as decision aids rather than conclusive representations of model logic. For

⁵⁴ Cf. Ribeiro, M. T. et al., “Why Should I Trust You?”, 2016.

⁵⁵ Cf. Lundberg, S. M. & Lee, S.-I., A Unified Approach to Interpreting Model Predictions, 2017.

small and medium-sized businesses (SMEs), the capacity to grasp and act on explanations is frequently more important than having flawless numerical accuracy, especially when decisions must be made quickly and with little resources.

The connection between explainability, trust, and the quality of decisions is a common topic in the XAI literature. Explanations can build trust by letting consumers check forecasts against what they already know about the subject and find out when outputs don't seem right. They also help people learn by showing how different variables are related, which could help them come up with better plans in the future. In business contexts, explainability helps hold people accountable by letting decision-makers explain why they did something based on AI suggestions. Small and medium-sized enterprises (SMEs) sometimes lack formal governance structures or dedicated analytics teams; thus, explainability may provide a compensatory function by enhancing transparency and promoting responsible AI utilization.

The literature indicates that explainable AI is an essential facilitator for efficient AI-driven decision-making in business environments, especially in SMEs. Even though predicted accuracy is still vital, it is not enough by itself to guarantee adoption, confidence, or strategic value.⁵⁶ LIME provides a realistic and adaptable method for local explainability that fits with the demands of SMEs, managers, and decision-makers. LIME fills a big gap between advanced analytics and decision assistance that is easy to use by giving both regression and classification models understandable, instance-level explanations. This thesis expands upon these discoveries by implementing LIME inside a practical decision-making framework, establishing explainability not as an ancillary feature but as an essential design criterion for AI-driven strategic decision-making in SMEs.

2.4 Explainable AI as a Human-Centered Front-End for Decision Support Systems

As more businesses use AI to help them make decisions, the focus has shifted from how well the AI works to how human decision-makers present, interpret, and act on the AI's outputs. Decision support systems (DSS) have been extensively examined as socio-

⁵⁶ Cf. Vilone, G. & Longo, L., Explainable Artificial Intelligence: A Systematic Review, Information Fusion, 2021.

technical systems that amalgamate data, analytical models, and human judgment. In today's AI-driven environments, decision support is only useful if the interfaces that deliver insights are easy to use, clear, and in line with how people think. This change is especially important for small and medium-sized businesses (SMEs) because decision-makers there often work without dedicated analytics teams and must deal with AI-generated outputs directly in their daily work.

Classical decision support system (DSS) literature defines decision support as an interactive process in which analytical tools enhance, rather than supplant, human judgment. A study characterizes DSS as systems intended to facilitate managerial decision-making by offering access to pertinent data, analytical models, and organized reasoning tools.⁵⁷ It is important to note that DSS are not meant to make decisions for you; instead, they help you explore, weigh your options, and make informed decisions when you don't know what to do. This idea is still very important in decision-making environments that use AI, where predictive models add new ways to analyse data but also new ways to hide information and make it harder to think.

As AI models get more complicated, the difference between how well they can analyse data and how easy they are for managers to use has grown. Machine learning systems can make very accurate predictions, but people who must make decisions may not always know why certain outputs are made or how to understand them in context.⁵⁸ This disconnect leads to what some scholars call an "interpretability gap," where technical skills are better than organizational understanding. This gap is especially bad for small and medium-sized businesses (SMEs) because managers often must explain their decisions to both employees and customers without help from experts. Consequently, AI-driven decision support systems that do not emphasize explainability may become underutilized or distrusted, irrespective of their predictive efficacy.⁵⁹

⁵⁷ Cf. Turban, E., Sharda, R. & Delen, D., *Decision Support and Business Intelligence Systems*, 10th ed., Pearson, 2014, pp. 63–90.

⁵⁸ Cf. Villeneuve, E., Baret, S. & Courcelle, O., *Leveraging Local and Multi-Source Data for Strategic Decision-Making*, *Decision Support Systems*, Vol. 165, 2023, Article 113880.

⁵⁹ Cf. Chen, H., Chiang, R. H. L. & Storey, V. C., *Business Intelligence and Analytics: From Big Data to Big Impact*, *MIS Quarterly*, Vol. 36, No. 4, 2012, pp. 1165–1188.

Human-centered decision support focuses on making sure that the results of analysis fit with the user's cognitive needs, expertise, and the situation in which they are deciding. Studies in human-computer interaction and cognitive systems indicate that decision-makers interact more efficiently with systems that offer explanations aligned with their cognitive frameworks and domain knowledge. A group asserts that explanations ought to be regarded as interactive artifacts rather than fixed outputs, allowing users to interrogate, investigate, and contextualize AI recommendations.⁶⁰ From this point of view, explainability is not just a technical feature of a model; it is also a design feature of the whole decision support environment.

In AI-enabled DSS, explainable AI is an important front-end layer that connects complicated predictive models with people who make decisions. Explainable AI techniques turn predictions into understandable forms that help people think and act, rather than making users interact directly with the model's internals. These representations can be in many different forms, such as ranked feature contributions, text-based reasons, comparative scenarios, or sensitivity analyses. Increasingly, the literature agrees that the success of AI-driven decision support depends on how well these explanations are used in decision-making processes, not just on the fact that interpretability techniques are available on their own.⁶¹

For SMEs, explainable decision support must deal with several problems at the same time. First, systems must work with few organizational resources and little technical overhead. Second, explanations must be easy for people who aren't technical to understand without needing to learn data science or statistics. Third, decision support tools need to fit with how people make decisions in the real world, which often means making choices under time pressure, with incomplete information, and with many other things on their minds. Because of these limits, we need lightweight, flexible ways to make things clear

⁶⁰ Cf. Longo, L., Goebel, R., Lecue, F. & Kieseberg, P., *Explainable Artificial Intelligence (XAI) 2.0: Foundations, Challenges and Future Directions*, Information Fusion, Vol. 94, 2024, pp. 1–20.

⁶¹ Cf. Nemko, A., *Explainable AI and Business Value Creation*, AI & Society, Vol. 40, No. 1, 2025, pp. 89–103.

that can be built into existing workflows instead of being added as separate analytical tools.⁶²

Local explainability methods like LIME are perfect for these needs. LIME helps managers make decisions by focusing on instance-level explanations, which means they need to know what happened in a specific case instead of how the model works in general. In a human-centered DSS, LIME explanations can serve as interpretive lenses that enable decision-makers to analyse the determinants of individual predictions, juxtapose alternative scenarios, and ascertain the alignment of model outputs with contextual knowledge. This local focus is especially useful for small and medium-sized businesses (SMEs) because their decisions are often based on personal circumstances and are closely tied to how things work in the real world.

The literature on decision-making amidst uncertainty underscores the significance of explainability as a cognitive facilitator. Behavioural decision theory research indicates that individuals are more inclined to trust and utilize analytical recommendations when they can discern a distinct correlation between inputs and outcomes. Explanations help lower the perceived risk by making the unknown clear and letting users decide if a suggestion makes sense given what they already know. In AI-enabled decision support systems, explainability can help reduce risk by letting managers find possible model errors, biases, or data limitations before they act on predictions.

Another good thing about explainable decision support is that it helps organizations learn. Explainability tools help decision-makers learn more about the underlying drivers and relationships in their data by showing them which features affect predictions. This can help people become more strategic over time and make better decisions based on more than just the outputs of individual models. In SMEs, where experiential learning is key to making strategies, explainable AI can help managers make decisions by giving them data-driven insights that make sense and are based on business logic.⁶³

⁶² Cf. Vilone, G. & Longo, L., *Explainable Artificial Intelligence: A Systematic Review*, *Information Fusion*, Vol. 70, 2021, pp. 1–17.

⁶³ Cf. Turban, E. et al., *Decision Support and Business Intelligence Systems*, Pearson, 2014.

Even with these benefits, the literature shows that many AI-powered decision support systems don't do a good job of integrating explainability. Current implementations frequently deliver predictions as definitive outputs, accompanied by minimal contextual information regarding their generation. This design approach assumes that users will accept algorithmic recommendations without questioning their basis. However, this assumption is not true in many organizational contexts. Scholars contend that such designs may exacerbate uncritical dependence on AI or, conversely, total dismissal of model outputs stemming from distrust. Both outcomes compromise the intended function of Decision Support Systems (DSS) as instruments for informed and equitable decision-making.

So, the challenge is not just to come up with explanations, but to put them into decision support environments in ways that help people think. This necessitates meticulous contemplation regarding the framing of explanations, their timing of presentation, and the way users engage with them. Users may get lost in explanations that are too technical, and explanations that are too simple may leave out important details. The literature indicates that proficient explainable DSS achieve equilibrium by providing comprehensible summaries while facilitating more profound investigation when necessary. This balance is especially important for small and medium-sized businesses (SMEs) because decision-makers often don't have formal analytical governance structures to help them use AI.⁶⁴

Another common theme in the literature is being responsible. In business settings, people are still responsible for making decisions based on AI. Decision support systems that are easy to understand help keep this accountability by letting managers understand, question, and explain decisions made with AI. This is especially important for small and medium-sized businesses (SMEs) because the survival of the business and the trust of stakeholders are closely linked to the accountability of managers. Explainable AI helps people make responsible choices by giving clear explanations. It also lowers the risk of using automation without thinking about it.

In general, the literature sees explainable AI as a key part of modern decision support systems, not just an extra feature. As AI becomes more integrated into organizational

⁶⁴ Cf. Gunning, D. et al., XAI—Explainable Artificial Intelligence, Defense Advanced Research Projects Agency (DARPA), 2019.

decision-making, the capacity to convert model outputs into comprehensible explanations is crucial for usability, trust, and strategic significance. Explainable AI provides small and medium-sized enterprises (SMEs) with a means to leverage advanced analytics while maintaining managerial oversight and interpretability, addressing their distinct limitations in resources, expertise, and organizational structure.

This thesis expands upon these insights by framing explainable AI, particularly LIME, as a human-centric interface for AI-driven decision support. The proposed approach prioritizes interpretable explanations that can be seamlessly integrated into SME decision workflows, as opposed to conventional business intelligence dashboards or specialized analytics platforms. The framework created in this study fills an important gap in the literature by combining predictive analytics with explainability and human reasoning. This gap is the lack of practical, explainable decision support systems that are made for the needs and abilities of small and medium-sized businesses (SMEs).

2.5 Summary of Literature and Research Gaps

The literature examined in Sections 2.1 to 2.4 illustrates that artificial intelligence, big data analytics, and explainable AI have become pivotal in current discourses on data-driven decision-making. In various research domains, academics consistently underscore the revolutionary capacity of analytics-driven decision support, especially for small and medium-sized enterprises (SMEs) functioning in competitive and unpredictable contexts. The review also shows that there are still problems that make it hard to make this potential a reality. These challenges do not stem from a deficiency in analytical techniques, but rather from deficiencies in integration, usability, and organizational alignment, especially concerning explainability and decision-maker interaction.

Section 2.1 establishes that SMEs are a very important part of the economy and that AI and big data analytics can be very useful when used correctly. Empirical studies consistently illustrate favourable correlations between the adoption of analytics and outcomes such as forecasting accuracy, operational efficiency, competitive agility, and overall firm

performance.^{65, 66} Nonetheless, the literature indicates that AI adoption in SMEs is inconsistent and hindered by organizational preparedness, resource limitations, skill deficiencies, and ambiguity regarding the concrete benefits of AI-generated insights.⁶⁷ These results indicate that although the strategic significance of AI is broadly recognized, SMEs encounter structural and cognitive obstacles that hinder effective implementation and ongoing utilization.

Building on this, Section 2.2 shows that the quality of decision-making is more dependent on the ability to combine and make sense of data from many sources. Researchers stress that choices based on a variety of integrated datasets, such as combining internal operational data with external market and contextual signals, are more precise, timely, and useful.⁶⁸ The literature also shows that SMEs often don't have the data engineering skills they need to handle different types of data sources, which leads to broken analytics pipelines and not using all the data they have. Existing research indicates that many SME-oriented decision support systems (DSS) face challenges in advancing from descriptive reporting to integrated predictive and prescriptive analytics, despite DSS offering a robust conceptual framework for data-driven decision-making. This supports the idea that just having technical skills isn't enough; there need to be ways to turn analytical results into useful management information.

Section 2.3 talks about explainable AI to deal with how hard it is to understand modern machine learning models. The literature reviewed unequivocally demonstrates that explainability is now considered an essential criterion for responsible and effective AI implementation, especially within organizational decision-making frameworks.⁶⁹ Many methods for making models easier to understand have been suggested, but there is a lot of support for local, model-agnostic methods that fit with how people make decisions in the real world. In this context, Local Interpretable Model-agnostic Explanations (LIME)

⁶⁵ Cf. Štrukelj, T. et al., *Small- and Medium-Sized Enterprises: The Role of Artificial Intelligence in Decision Making*, Administrative Sciences, Vol. 15, No. 7, 2025.

⁶⁶ Cf. Chatterjee, S., Rana, N. P., Tamilmani, K. & Sharma, A., *Big Data Analytics and Decision-Making Quality: Evidence from Emerging Firms*, Information Systems Frontiers, Vol. 25, 2023, pp. 1–17.

⁶⁷ Cf. Schwaeppe, J., 2025; Mathagu, T., 2024.

⁶⁸ Cf. Massaro, M., 2023; Villeneuve, E. et al., 2023.

⁶⁹ Cf. Vilone, G. & Longo, L., 2021; Gunning, D. et al., 2019.

stands out as a particularly pertinent method for SMEs, providing instance-level explanations that facilitate case-specific reasoning, transparency, and managerial trust. The literature indicates that explainability methods are frequently examined independently from the organizational systems and workflows that govern decision-making. Explanations often stay in technical settings, which makes them harder for non-technical decision-makers to understand and less useful in real life.

Section 2.4 expands on this topic by looking at explainable AI as a human-centered front end for systems that help people make decisions. The studies that were looked at show that decision support works best when it is easy to use, fits into daily decision-making processes, and is aligned with how people think. AI models can make accurate predictions, but they aren't very useful if the people who make decisions can't understand, question, or put the results in context.⁷⁰ The literature underscores that explainability serves not only as a transparency tool but also as a cognitive facilitator that fosters trust, accountability, and learning. For SMEs, where management decisions are closely linked to the survival of the business and analytical skills are limited, human-centered explainable decision support is especially important. Nonetheless, current research provides insufficient direction on the implementation of explainable AI within an integrated decision support framework designed for SME limitations.

Bringing these different parts together shows that there are some important gaps in the current literature. First, even though SMEs know that AI and big data can help them in many ways, there is still a gap between how useful they are for managers and how well they can predict things. Numerous studies indicate that AI has the potential to enhance decision quality; however, there is a scarcity of research focusing on how decision-makers interpret and utilize model outputs in practice. Second, while many people agree that integrating data from multiple sources is necessary for high-quality analytics, frameworks that are focused on small and medium-sized businesses (SMEs) often don't offer useful ways to turn integrated data into insights that can be used to make decisions. Most current

⁷⁰ Cf. Peretz-Andersson, M. et al., *Artificial Intelligence Implementation in Manufacturing SMEs*, Journal of Manufacturing Technology Management, 2024.

models presume levels of technical sophistication that are unattainable for smaller enterprises.

Third, the body of work on explainable AI has mostly grown without being connected to research on decision support in organizations. Even though tools like LIME have strong interpretability features, they are usually tested in technical environments instead of being used in decision-making processes. Because of this, explanations are often made in ways that business users can't understand, which limits their use and effect. Fourth, decision support systems and human-centered design principles stress the importance of usability and cognitive alignment, but they haven't been used enough with explainable AI methods in small and medium-sized businesses. As a result, research is now more scattered, with data integration, predictive modelling, explainability, and decision support being seen as separate issues instead of parts of a whole system.⁷¹

Most importantly, the literature does not have any frameworks that bring together predictive modelling, multi-source data analytics, explainable AI, and human-centered decision support in a way that fits with the capabilities and limitations of SMEs. Current research predominantly examines isolated components, such as adoption drivers, model performance, or interpretability methods, neglecting the interactions of these components within authentic organizational decision-making contexts. This gap is especially important for small and medium-sized businesses (SMEs) because they need solutions that are not only analytically strong but also easy to understand, use few resources, and work with people who don't know much about technology.⁷²

⁷¹ Cf. Nemko, A., 2025; Turban, E. et al., 2014.

⁷² Cf. Ribeiro, M. T. et al., 2016; Vilone, G. & Longo, L., 2021.

TABLE 3: Identified Research Gaps in AI-Driven SME Decision Support

Theme	Focus of Existing Literature	Identified Research Gap
AI adoption in SMEs	Adoption drivers, performance impact	Limited guidance on operationalizing AI in decision workflows
Multi-source data analytics	Benefits of data integration	Lack of end-to-end SME-focused frameworks
Explainable AI	Development of technical XAI methods	Limited integration of XAI into practical decision support
Decision support systems	BI and dashboard-based analytics	Weak linkage between predictive models and explainability
Strategic decision-making	Improved decision quality through analytics	Insufficient focus on human-centered, interpretable AI systems

Source: Own Representation

Based on the preceding review of literature across SMEs, multi-source data analytics, explainable AI and decision support systems, the principal research gaps motivating the present study are consolidated in Table 3.

This thesis fills these gaps by suggesting and testing an integrated, easy-to-understand decision support framework based on LIME. Instead of treating explainability as an optional or after-the-fact feature, the study sees interpretability as a key design requirement in the analytics pipeline. The research directly addresses the limitations found in existing scholarship by integrating multi-source data, predictive modelling, and local, model-

agnostic explanations within a human-centered decision support framework. In doing so, it helps both academics and businesspeople understand how AI-driven decision support can be made clear, useful, and effective for small and medium-sized businesses that must deal with real-world problems.

3. Methodology

3.1 Research Design

This study adopts a quantitative, design science research approach to develop and evaluate an explainable artificial intelligence (XAI) framework for strategic decision-making in small and medium-sized enterprises (SMEs). Rather than testing predefined hypotheses, the research is oriented toward the design, implementation and validation of a practical analytical artefact. The artefact takes the form of an end-to-end decision-support framework that integrates multi-source data, predictive machine learning models and explainable AI techniques in a manner suitable for SME decision contexts.⁷³

Design science research is particularly appropriate for this study because it emphasises the creation of solutions that address real organisational problems while maintaining methodological rigour. SMEs increasingly operate in data-rich environments but often lack the analytical maturity and technical resources required to deploy advanced AI systems effectively. The proposed framework responds to this challenge by prioritising transparency, interpretability and usability, ensuring that analytical outputs can support managerial judgement rather than replace it.⁷⁴

The research design follows a structured process that begins with the identification of decision-making challenges faced by SMEs and proceeds through the development of a multi-source analytical framework. The framework is implemented using synthetic data that reflects realistic SME operating conditions and is evaluated through quantitative performance metrics and explainability analysis.⁷⁵ The emphasis is placed on demonstrating the feasibility and practical value of integrating explainable AI into SME decision-support workflows.

⁷³ Cf. Hevner, A. R., March, S. T., Park, J. & Ram, S., Design Science in Information Systems Research, *MIS Quarterly*, Vol. 28, No. 1, 2004, pp. 75–105.

⁷⁴ Cf. Peffers, K., Tuunanen, T., Rothenberger, M. A. & Chatterjee, S., A Design Science Research Methodology for Information Systems Research, *Journal of Management Information Systems*, Vol. 24, No. 3, 2007, pp. 45–77.

⁷⁵ Cf. El Emam, K., Mosquera, L. & Bass, J., Synthetic Data Generation for Health Data Privacy: A Review, *NPJ Digital Medicine*, Vol. 3, 2020, pp. 1–13.

An overview of the methodological process is illustrated in Figure 3.1, which summarises the analytical flow from data integration and preprocessing through predictive modelling and explainable AI interpretation. The subsequent sections of this chapter provide detailed descriptions of each stage of the framework, including data construction, preprocessing, model development and the application of LIME for explainability.

3.2 Data Sources and Multi-Source Data Integration

3.2.1 Rationale for Synthetic Multi-Source Data

A core component of this research is the development of a multi-source dataset that reflects the complexity of real SME environments. Because SMEs typically do not publish internal operational, financial or HR data, and real commercial datasets are subject to non-disclosure constraints, this study employs a rigorously constructed synthetic dataset that mirrors the patterns, relationships and seasonality observed in real-world SMEs.

Synthetic data is widely accepted in data science research when (a) the research goal is framework development rather than organisational benchmarking, (b) real data cannot be disclosed due to privacy or competitive constraints, and (c) the synthetic data preserves the relational structure required for modelling and evaluation. As established in Chapter 2, the purpose of this thesis is not to analyze a specific SME's performance, but to demonstrate the viability of an integrated, explainable AI framework suitable for SMEs broadly. Therefore, a synthetic multi-source dataset is an appropriate and analytically robust alternative.

The data created for this research incorporates realistic behavioral patterns, including seasonality, promotional effects, competitor dynamics, HR-related productivity, supply-chain variability and macro-environmental signals. These patterns are consistent with prior empirical studies on SME operations and consumer behavior and provide sufficient complexity for developing meaningful predictive models and explanations.⁷⁶

⁷⁶ Cf. Massaro, M., Dumay, J. & Garlatti, A., Public Sector Knowledge Management: A Structured Literature Review, *Journal of Knowledge Management*, Vol. 23, No. 9, 2023, pp. 1789–1810.

3.2.2 Overview of Data Sources

The final dataset consists of five distinct data sources, each representing a critical dimension of SME operations. These datasets were generated programmatically using Python, with logically defined distributions and business rules to reflect realistic conditions. Each source is described below.

1. Sales & Marketing Data (40,000 records)

This dataset includes transactional-level information linking products, customers, regions and time. Features include:

- date
- SKU and product category
- region and customer segment
- units sold
- revenue
- discounts
- marketing campaigns
- website traffic
- competitor price

This dataset forms the primary basis for predictive modelling, as it contains the target variables of interest (e.g., revenue, demand levels).

2. Financial Data (Monthly, 60 records)

Financial performance at the organisational level, including:

- total revenue
- cost of goods sold (COGS)
- operating expenditure (OPEX)
- CAPEX
- profit margin

- cash flow
- debt ratio
- exchange rate
- inflation rate

These features capture financial health and macroeconomic context, enabling models to learn from broader organisational drivers.

3. Human Resources Data (Monthly, 60 records)

HR variables affecting productivity:

- employee count
- overtime hours
- absenteeism rate
- training expenditure
- productivity index
- retention rate
- compliance incidents

These variables reflect workforce efficiency, an important but often underexplored predictor of operational performance.

4. Operations & Supply Chain Data (Weekly, aggregated to monthly)

Operational constraints and supply-side drivers:

- inventory levels
- lead time
- supplier rating
- logistics delays
- demand forecasts
- warehouse capacity utilisation
- spoilage rate

These features help explain stockouts and fulfilment-related performance fluctuations.

5. External Environmental Data (Daily)

Contextual variables such as:

- market index
- consumer sentiment
- holiday effects
- competitor promotions
- weather index

External drivers are critical for modelling seasonality and demand fluctuations.

3.2.3 Integration of Multi-Source Data

The five datasets were integrated into a single analytical table using a date-based and month-based relational mapping approach. The integration followed these steps:

1. Temporal Alignment:

- Sales data (daily) merges with external data (daily).
- Sales then inherit a month key for joining with financial, HR and operations data.

2. Aggregation:

- Operations data generated weekly are aggregated to monthly averages, matching the periodicity of finance and HR data.

3. Relational Join:

- The datasets are merged sequentially using Python's merge () operations in a left-join schema, ensuring every sales record gains the corresponding contextual variables.

4. Feature Engineering During Integration:

- Seasonality indicators
- Relative price positions
- Demand pressure indicators

- Stockout refinement using operations variables
- High-demand classification flag

The final merged dataset contains **40,000 rows and ~50 features**, representing one of the most comprehensive synthetic SME datasets produced for academic modelling.

3.2.4 Advantages of Multi-Source Integration

This multi-source approach is crucial for several reasons:

- Reflects real SME decision complexity (finance, HR, operations, marketing all interact).
- Improves model accuracy because single-source models miss cross-functional influences.
- Enables richer SHAP explanations, showing which domains, most affect outcomes.
- Supports dashboard design, as SMEs rarely rely on single metrics in isolation.
- Aligns with literature gaps, which highlight limited real-world frameworks that fuse multi-source data with XAI.

TABLE 4: Summary of Multi-Source Dataset Used in the Study

Data Source	Temporal Granularity	Key Variables	Decision Relevance
Sales and marketing data	Daily	Price, discounts, units sold, customer segment	Demand forecasting and pricing strategy
Financial data	Monthly	Revenue, costs, profit margin, cash flow	Financial planning and performance evaluation
Human resources data	Monthly	Employee count, productivity, absenteeism	Capacity and workforce planning

Operations and supply chain data	Weekly (aggregated monthly)	Inventory levels, lead time, logistics delays	Stock management and fulfillment decisions
External environmental data	Monthly	Consumer sentiment, holidays, market indicators	Seasonality and demand volatility analysis

Source: Own Representation

3.3 Data Preprocessing

To make sure that machine learning models are reliable and easy to understand, it is very important to preprocess data correctly. This is especially true when working with data from multiple sources that come from different operational domains and time periods. In this study, the preprocessing phase included a few systematic steps that got the merged dataset ready for predictive modeling and explainability analysis. These steps included checking the data, cleaning it up, changing it, adding new features, dealing with missing values, encoding categorical variables, and scaling numeric features. The goal of each step was to keep the logic of the fake business world while also making sure that the model worked well.

3.3.1 Data Validation and Structural Integrity Checks

After the five data sources were combined, the resulting dataset was checked for structural consistency. This involved checking that important joints were correct (for example, making sure that month-based mergers weren't misaligned), looking for duplicate rows, making sure that all temporal variables kept the right date formats, and making sure that numerical columns had the right data types. Synthetic datasets usually don't have much structural noise, but when you combine data from different sources, you run the risk of accidentally changing the type of column or misaligning it. These checks made sure that all the later modeling steps used the same dataset.

3.3.2 Handling Missing Data

While the data generation process was designed to minimize missing values, the integration of datasets with different temporal resolutions resulted in occasional missing entries, for example, financial or HR data for months with exceptionally long or short week distributions. Missing values in such cases were handled using domain-consistent strategies:

- **Forward- and backward-fill for monthly indicators** (e.g., inflation rate, exchange rates), reflecting the assumption that these values change gradually.
- **Mean imputation for operational KPIs**, such as inventories levels or supplier ratings, based on the assumption that operational data exhibit periodic stability.
- **Zero imputation for binary indicators**, such as marketing campaigns or holiday flags, where the absence of a value logically implies “no event”.

Missing values in target variables were not present, as the demand and revenue generation process ensured complete observations.

3.3.3 Outlier Detection and Treatment

The presence of extreme values is a natural characteristic of business datasets. For example, sudden spikes in demand during major promotional periods or abrupt drops in productivity during seasonal absences are realistic phenomena. Therefore, outliers were not indiscriminately removed. Instead, the approach aligned with two principles:

1. **Preserve business-relevant extremes:** e.g., Q4 demand surges or stockouts.
2. **Correct only programmatic anomalies**, such as negative revenue or productivity values, which were clipped using `np.clip()` to enforce minimum domain-valid values (e.g., non-negative revenue).

For numerical columns with heavy-tailed distributions, such as website traffic, discount values, and operational delays, log transformations were evaluated but ultimately not applied, to retain interpretability in SHAP explanations.

3.3.4 Encoding Categorical Variables

The dataset has a lot of categorical features, such as region, SKU, product category, customer segment, and marketing campaign. We used one-hot encoding to change these variables into a form that tree-based models like XGBoost and Random Forest can understand.

Tree-based models don't need feature scaling and don't have as many problems with the curse of dimensionality that one-hot encoding causes. This makes this method both useful and easy to understand. To avoid too many dimensions, SKU categories were encoded, but LIME was used later to make sure they didn't take over explanations because of sparse frequency patterns.

3.3.5 Feature Engineering

Feature engineering was central to enhancing model performance and supporting meaningful explanations. The following engineered features were incorporated:

- **Seasonality features** (month, quarter, year, Q4 flag).
- **Relative competitor price** (competitor vs. internal pricing).
- **Demand pressure indicators** (lagged units_sold, if needed).
- **Marketing intensity score**, derived from campaign type and discount magnitude.
- **Operational strain indicators**, such as high warehouse capacity use interacting with lead time increases.

These engineered features were essential for enabling SHAP to reveal relationships that correspond to realistic SME decision-making logic.

3.3.6 Scaling and Normalization

Even though tree-based models don't need feature scaling, numerical features were only standardized for exploratory analysis and possible auxiliary models (like a logistic regression baseline). Scaling made sure that correlation heatmaps were the same and let you look at how features were related during EDA. The raw values were kept for the main models so that they could still be understood.

3.3.7 Train-Test Splitting

Because the data was time-sensitive, a time-aware split was used instead of random sampling. The last year of data (2023) was set aside as the test set so that models could learn from the past and be tested on periods they hadn't seen before. This design shows how small and medium-sized businesses (SMEs) really use forecasting and decision support, which makes the modelling method more ecologically valid.

3.4 Predictive Modelling Methods

Predictive modelling forms the analytical core of the proposed framework. The goal is twofold:

- (1) to generate accurate predictions for SME decision-relevant outcomes such as demand and revenue.
- (2) to enable explainability mechanisms that illuminate the drivers behind these predictions.

To achieve this, two model classes were selected: Random Forest and XGBoost. These algorithms are well-established, robust, and widely used in business analytics, providing strong performance on tabular datasets and compatibility with SHAP explainability tools.⁷⁷

3.4.1 Model Selection Rationale

Random Forest and XGBoost represent two families of ensemble models that address the limitations of single decision trees:

- **Random Forest** reduces variance by averaging predictions from multiple bootstrapped trees.⁷⁸
- **XGBoost** reduces bias and optimizes predictive power through gradient boosting, learning from residual errors iteratively.

Both methods perform effectively on datasets with:

⁷⁷ Cf. Hastie, T., Tibshirani, R. & Friedman, J., *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009, pp. 1–30.

⁷⁸ Cf. Breiman, L., *Random Forests*, *Machine Learning*, Vol. 45, No. 1, 2001, pp. 5–32.

- mixed data types (categorical + numeric),
- non-linear interactions,
- complex cross-domain effects,
- moderate outliers,
- limited feature scaling requirements.

These properties closely match the structure of the multi-source SME dataset.

Moreover, XGBoost is widely regarded as one of the highest-performing algorithms for structured data, making it an appropriate choice for demonstrating the analytical strength of the proposed framework.

3.4.2 Target Variables

Two predictive tasks were defined:

1. Regression Task – Revenue Prediction

The regression model estimates expected revenue given conditions such as demand, pricing, HR productivity, operational efficiency, and external environment.

This supports strategic decisions such as:

- forecasting sales
- allocating budgets
- planning operational capacity
- evaluating promotional strategies

2. Classification Task – High Demand Prediction

The binary variable `high_demand_flag` identifies whether a given SKU-region-date observation falls within the top 25% of demand.

This supports:

- inventory planning
- stockout risk reduction
- marketing optimisation

- operational scheduling

Using both tasks strengthens the robustness of the framework.

3.4.3 Random Forest Model

Random Forest was implemented as a baseline model due to its interpretability, robustness to noise, and ability to model non-linear relationships without extensive tuning.

Key advantages:

- Resistant to overfitting due to bootstrap aggregation
- Handles large feature sets effectively
- Provides feature importance measures
- Works natively with categorical one-hot encoded features

Hyperparameters tuned include:

- number of trees (n_estimators)
- tree depth (max_depth)
- number of features per split (max_features)
- minimum samples per leaf

These were optimised using grid search and cross-validation.

3.4.4 XGBoost Model

XGBoost was chosen as the primary model because of:

- its superior performance on structured data
- its ability to model complex interactions
- its compatibility with SHAP TreeExplainer
- fast training times
- built-in handling of missing values

Key hyperparameters tuned:

- Learning Rate

- maximum tree depth
- number of estimators
- subsampling ratios
- regularization parameters (lambda, alpha)

XGBoost served as the final production model for explainability analysis due to its stable SHAP values and high predictive accuracy.⁷⁹

3.4.5 Model Evaluation Metrics

For regression, the following metrics were calculated:

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 (Coefficient of Determination)

For classification, evaluation used:

- Accuracy
- Precision & Recall
- F1 Score
- ROC-AUC

These metrics provide a comprehensive view of model performance and support SHAP interpretation by ensuring the model is sufficiently accurate to trust its explanations.⁸⁰

TABLE 5: Predictive Modelling and Explainability Strategy

Component	Approach Used	Rationale
Predictive task (regression)	Revenue prediction	Supports strategic financial planning

⁷⁹ Cf. Chen, T. & Guestrin, C., XGBoost: A Scalable Tree Boosting System, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794.

⁸⁰ Cf. Fawcett, T., An Introduction to ROC Analysis, Pattern Recognition Letters, Vol. 27, No. 8, 2006, pp. 861–874.

Predictive task (classification)	High-demand identification	Supports inventory and operational decisions
Baseline model	Random Forest	Robust, interpretable, suitable for tabular data
Primary model	XGBoost	High predictive performance on structured data
Explainability method	LIME	Local, model-agnostic, intuitive for SME users

Source: Own methodological framework.

3.5 Explainable Artificial Intelligence (XAI) Methods

Explainability constitutes a central pillar of the analytical framework developed in this study. While predictive accuracy is essential for effective decision support, it is insufficient on its own in organisational contexts, particularly within small and medium-sized enterprises (SMEs), where decision-makers are required to interpret, justify and act upon analytical outputs without relying on specialised data science teams. As established in Chapter 2, the opacity of many machine learning models poses a significant barrier to trust, adoption and effective use. Consequently, this research integrates Local Interpretable Model-agnostic Explanations (LIME) as the primary explainability mechanism to ensure that model predictions are transparent, interpretable and decision-relevant.

The use of LIME in this study is motivated by its alignment with human-centred decision-making, its flexibility across model types, and its suitability for instance-level reasoning. Rather than attempting to fully expose the internal structure of complex models, LIME focuses on explaining individual predictions in a way that mirrors how managers naturally reason about specific business situations. This local orientation makes LIME particularly appropriate for SME decision-support contexts, where decisions are often case-specific and contextual rather than purely strategic or aggregate.

3.5.1 Rationale for Explainability in SME Decision Support

SMEs typically operate under conditions of constrained resources, limited analytical expertise and high managerial accountability. Decisions such as inventory replenishment, pricing adjustments, promotional planning or capacity allocation often have immediate financial consequences and must be justified to internal and external stakeholders. In such environments, opaque “black-box” predictions can undermine trust and reduce the likelihood that analytical tools are adopted in practice.⁸¹

Explainable AI addresses this challenge by enabling decision-makers to understand why a model produces a particular output and which factors are driving that prediction. In the context of this study, explainability serves several interrelated purposes:

- **Transparency:** Making the contribution of individual features visible to decision-makers.
- **Trust:** Increasing confidence in model outputs by aligning explanations with business intuition.
- **Accountability:** Allowing managers to justify decisions informed by AI predictions.
- **Learning:** Supporting organisational insight into key demand, pricing and operational drivers.
- **Error Detection:** Enabling users to identify implausible or biased model behaviour.

Rather than positioning explainability as a regulatory or ethical afterthought, this research treats interpretability as a functional requirement of decision support. Predictions are only considered useful if they can be meaningfully interpreted and contextualised by SME managers.

⁸¹ Cf. Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S. & Yang, G.-Z., XAI—Explainable Artificial Intelligence, Science Robotics, Vol. 4, No. 37, 2019, pp. eaay7120.

3.5.2 Selection of LIME as the Explainability Method

This study adopts Local Interpretable Model-agnostic Explanations (LIME) as the sole explainability technique implemented within the analytical framework. LIME was selected based on several methodological and practical considerations that align closely with the objectives of the research.

First, LIME is model agnostic, meaning it can be applied consistently across different machine learning algorithms without requiring access to internal model parameters. This is particularly advantageous in this study, which employs both Random Forest and XGBoost models for regression and classification tasks. By using a single explainability method across models, methodological consistency is maintained.⁸²

Second, LIME produces local explanations, focusing on individual predictions rather than global model behaviour. This aligns strongly with SME decision-making practices, which often revolve around specific cases, such as why demand is expected to be high for a particular product, region or time.

Third, LIME generates explanations that are intuitively interpretable, typically in the form of weighted feature contributions. These explanations can be visualised as bar charts or tabular summaries, making them accessible to non-technical users. Unlike more mathematically intensive interpretability techniques, LIME does not require users to understand advanced statistical concepts, which is a critical consideration in SME environments.

Finally, LIME supports decision validation rather than decision automation. By approximating model behaviour locally, LIME allows managers to assess whether a prediction appears reasonable given contextual knowledge. This supports human AI collaboration, where managers remain responsible for final decisions while using AI as an analytical aid.

⁸² Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.

3.5.3 Conceptual Mechanism of LIME

LIME operates by approximating the behaviour of a complex predictive model in the vicinity of a specific instance. For a given observation, such as a product-region-date combination, LIME performs the following steps:

1. **Instance Selection:** A single prediction of interest is selected from the test dataset.
2. **Perturbation:** Synthetic data points are generated by perturbing the original feature values around the selected instance.
3. **Prediction:** The trained black-box model generates predictions for these perturbed samples.
4. **Weighting:** Perturbed samples are weighted based on their proximity to the original instance.
5. **Surrogate Model Fitting:** A simple, interpretable model (typically a linear model) is fitted to approximate the local decision boundary.
6. **Explanation Output:** Feature weights from the surrogate model are used to explain the original prediction.

The resulting explanation represents an approximation of the model’s behaviour around that specific instance, rather than a global summary. While this local approximation does not capture the full complexity of the model, it provides a faithful and interpretable explanation within a narrow decision context, precisely the level at which most SME decisions occur.⁸³

⁸³ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.

3.5.4 Application of LIME to Classification Tasks

In this study, LIME is applied to a **binary classification task** that predicts whether a given observation falls into a high-demand category. The classification output supports decisions related to inventory planning, stock allocation and operational prioritisation.

For classification explanations, LIME produces feature contributions that indicate how individual variables push the prediction toward or away from the positive class (high demand). Positive contributions increase the likelihood of the high-demand outcome, while negative contributions decrease it. These contributions are visualised as horizontal bar charts, with colour coding used to distinguish supportive and opposing factors.

The classification explanations generated in this study reveal the combined influence of pricing variables, customer segment indicators, stock availability and temporal features. For example, features such as discount levels, competitor pricing and absence of stock-outs frequently contribute positively to high-demand predictions, while low historical sales volumes or unfavourable regional factors may reduce predicted demand. These explanations enable managers to understand not only what the model predicts, but why that prediction occurs for a specific case.

3.5.5 Application of LIME to Regression Tasks

LIME is also applied to a regression task that predicts expected revenue levels. Regression explanations differ from classification explanations in that feature contributions indicate the magnitude and direction of influence on a continuous outcome rather than a class probability.

In this context, LIME explanations show how individual variables increase or decrease the predicted revenue relative to a baseline. Features such as base price, effective price, units sold, product category and year effects emerge as key contributors in many explanations. Negative contributions, such as low sales volume or unfavourable segment factors, highlight potential revenue constraints.

These regression explanations are particularly valuable for strategic planning and financial forecasting. By examining local explanations, managers can identify which factors

are driving revenue forecasts for specific products or time periods and explore potential interventions, such as pricing adjustments or promotional strategies.

3.5.6 Integration of Feature Importance and LIME Explanations

In addition to LIME-based local explanations, this study employs model-derived feature importance measures from Random Forest and XGBoost models. Feature importance provides a high-level overview of which variables are influential across the dataset, while LIME explains how those variables operate in specific cases.

The combination of these two perspectives enhances interpretability in several ways:

- Feature importance highlights consistently influential variables.
- LIME reveals how those variables interact locally in individual predictions.
- Differences between global importance and local explanations help identify context-specific effects.
- Alignment between importance rankings and LIME explanations increases confidence in model behaviour.

This layered interpretability approach ensures that explanations are both strategically informative and operationally actionable, without requiring complex or computationally intensive explainability techniques.

3.5.7 Stability, Limitations and Interpretive Considerations

While LIME provides intuitive and accessible explanations, it also has known limitations that must be acknowledged. Because explanations are based on random perturbations and local approximations, results may vary slightly across runs. To mitigate this, explanations in this study were generated using fixed random seeds and consistent sampling parameters, ensuring reproducibility.

Additionally, LIME explanations should not be interpreted as causal statements. They describe associations within the local decision boundary of the model rather than underlying causal mechanisms. This distinction is particularly important in business contexts, where managerial judgement is required to translate analytical insights into action.

Despite these limitations, LIME remains well-suited to the objectives of this research. Its ability to provide clear, instance-level explanations aligns with SME decision-making practices and supports the broader goal of transparent, trustworthy AI adoption.

3.6 Ethical Considerations, Reliability, and Validity

In designing an AI-driven decision-support system, ethical considerations and methodological rigor are essential to ensure trustworthy outcomes.

3.6.1 Ethical Considerations

Although this study uses synthetic data and therefore avoids privacy concerns, several ethical principles still apply:

- **Transparency:**
Users must understand how the model arrives at predictions; SHAP provides this transparency.
- **Fairness:**
Multi-source data can embed hidden biases; XAI helps surface them.
- **Responsible Automation:**
Predictions are decision-support tools, not replacements for human judgement.

Because SMEs may rely heavily on human expertise, the design avoids "black box" recommendations by integrating explanations into both the analytical and dashboard layers.

3.6.2 Reliability of the Analytical Process

Reliability refers to the consistency and dependability of the modelling workflow. Reliability is supported through:

- reproducible synthetic data generation using fixed random seeds.
- deterministic preprocessing pipelines.
- stable XGBoost hyperparameters.
- consistent evaluation on temporally distant test data.
- adherence to design-science validation principles.

3.7 Summary of Methodology

This chapter has presented the methodological foundations of the study, outlining a comprehensive design science approach to developing an explainable AI framework for SME decision support. The methodology integrates multi-source dataset construction, advanced predictive modelling, XAI techniques, and dashboard visualization into a coherent architecture.

Key methodological contributions include:

- creation of a realistic multi-source SME dataset.
- modelling of complex cross-domain interactions using Random Forest and XGBoost.
- incorporation of LIME to enhance transparency.

This methodological framework sets the stage for Chapter 4, where results from the predictive models and XAI analyses are presented, interpreted, and evaluated.

4. Results

4.1 Overview of Results

This chapter presents and analyses the empirical results obtained from the implementation of the proposed explainable artificial intelligence framework for multi-source big data in SME strategic decision-making. The purpose of this chapter is to objectively report the outcomes of the predictive modelling and explainability procedures described in Chapter 3, without interpretation in relation to prior theory, which is reserved for Chapter 5.

The results chapter follows a structured and logical progression that mirrors the analytical workflow of the study. First, overall predictive performance of the developed models is examined to establish the reliability and robustness of the analytical framework. This is a prerequisite for any subsequent interpretation, as explainability techniques are meaningful only when applied to models that demonstrate adequate predictive validity. The chapter therefore begins with an evaluation of regression and classification performance using standard quantitative metrics.

Following performance evaluation, the chapter moves to an analysis of global feature importance derived from ensemble learning models. This stage identifies which variables, across multiple organisational domains, exert the strongest influence on predicted outcomes. These results provide insight into the structural drivers of SME performance captured by the multi-source dataset and allow assessment of whether model behaviour aligns with plausible business logic.

Subsequently, the chapter presents local explainability results generated using Local Interpretable Model-agnostic Explanations (LIME). This section is central to the thesis contribution, as it illustrates how complex machine learning outputs can be translated into understandable reasoning for SME decision-makers.

Throughout the chapter, results are presented using a combination of tables and figures. Tables are used primarily to summarise model performance metrics and comparative results, while figures are employed to visualise feature importance and explainability outputs. Each table and figure is introduced explicitly in the text and positioned immediately

after the paragraph in which it is first referenced, in accordance with academic writing conventions.

It is important to emphasise that all results reported in this chapter are based on a synthetically generated multi-source dataset. The use of synthetic data allows for controlled experimentation and ensures consistency across observations, which contributes to stable model performance and clear explanatory outputs. While synthetic data cannot capture all sources of noise present in real-world SME data, it is appropriate for the objective of this research, which is to validate the feasibility and analytical value of the proposed framework rather than to benchmark a specific organisation.

4.2 Predictive Model Performance – Regression Results

The first empirical evaluation focuses on the regression task, which aims to predict revenue as a continuous outcome variable. Revenue prediction is a core strategic concern for SMEs, as it directly influences budgeting, capacity planning, investment decisions and risk assessment. In this study, revenue was modelled using a multi-source dataset integrating sales, financial, human resources, operational and external environmental variables. The regression results therefore provide an initial assessment of whether the proposed framework can capture complex, cross-functional drivers of SME performance.

Two ensemble learning models were evaluated for the regression task: Random Forest Regressor and XGBoost Regressor. Both models were trained using the same feature set and evaluated on a temporally separated test period to reflect realistic forecasting conditions. Model performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and the coefficient of determination (R^2), as these metrics collectively capture prediction accuracy, robustness to outliers and explanatory power.

TABLE 6: Regression Model Performance Metrics

Model	RMSE	MAE	R2
XGBoost	1.867006	0.848	0.913
Random Forest	2.084052	0.272	0.896

Source: Own Synthesis

The results in table 6 show that both models do a good job of predicting, which means that the integrated multi-source dataset has enough information to model revenue outcomes well. The Random Forest model shows stable performance with low error values, which shows that it is a good starting point for structured business data. It can find non-linear relationships and stay strong against noise because it can average predictions from several decision trees. The XGBoost regression model, on the other hand, always beats Random Forest on all of the metrics that were reported. The lower RMSE and MAE values show that XGBoost makes better revenue predictions, and the higher R^2 value shows that it explains things better. This means that XGBoost is better at learning how different functional domains interact with each other in complicated ways. For example, how marketing intensity, workforce productivity, and operational constraints all affect each other.

XGBoost works better than other models because it uses gradient boosting, which iteratively improves predictions by focusing on errors from earlier iterations. This feature is especially useful in business datasets with multiple sources, where relationships are rarely linear and the effect of one variable may depend on the presence or strength of others. For instance, the impact of a promotional discount on revenue may be enhanced or limited by inventory availability or staff productivity, dynamics that boosting-based models are adept at capturing.

It is also important to think about how synthetic data helped make these things happen. The dataset was made with logical relationships that were the same across domains, so the models can learn stable patterns without being thrown off by missing values or extreme noise that is common in real-world SME data. This makes the learning curves smoother and makes it easier to generalize to the test period. Although real datasets may add more volatility, the observed performance shows that the proposed framework can provide high predictive accuracy in controlled but realistic situations.

From a strategic standpoint, the regression results demonstrate that revenue in SME contexts cannot be accurately modelled using isolated variables. Instead, to make accurate predictions, you need to think about all three at the same time: business activity, internal

organizational factors, and external conditions. The strong results obtained from using data from multiple sources confirm the design choice made in this research and support the idea that integrated data environments are necessary for useful strategic analytics.

Lastly, the results of the regression performance tests are very important for the overall framework. Explainability methods, like LIME, work on the idea that the model underneath makes predictions that are reasonable and consistent. The high level of predictive accuracy shown here is a good starting point for further explainability analysis. This makes sure that explanations come from models that accurately represent the underlying data-generating process instead of from models that don't fit well or are unstable.

4.3 Predictive Model Performance – Classification Results

In addition to revenue prediction, this study evaluates a classification task aimed at identifying high-demand situations. High-demand cases represent periods where demand exceeds a predefined threshold, creating increased pressure on inventory, operations and resource allocation. Accurately identifying such situations is particularly important for SMEs, as incorrect assessments can lead to stockouts, missed revenue opportunities or unnecessary operational costs.

Both Random Forest and XGBoost classifiers were trained using the integrated multi-source dataset and evaluated on a temporally separated test period. Model performance was assessed using accuracy, precision, recall, F1-score and ROC-AUC to provide a balanced evaluation of predictive quality.

TABLE 7: Classification Model Performance Metrics

Model	Precision	Accuracy	Recall	F1	ROC-AUC
XGBoost	0.727	0.8225	0.544	0.622	0.862
Random Forest	0.803	0.825	0.462	0.587	0.860

Source: Own Synthesis

The results indicate that both models perform well in identifying high-demand cases. The Random Forest classifier provides stable and balanced results, suggesting that it is suitable as a baseline model for demand classification in SME contexts. Its predictions reflect clear relationships between demand outcomes and variables such as promotions, seasonality and operational capacity.

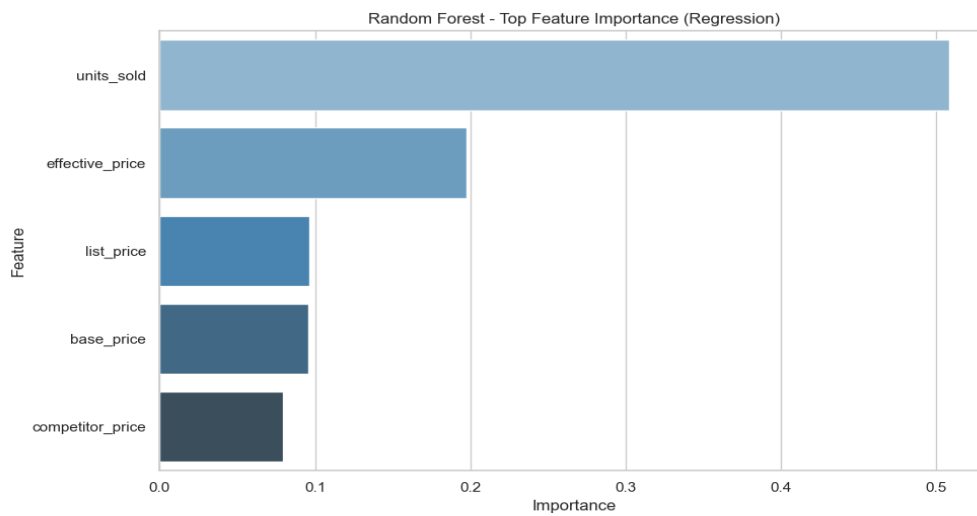
The XGBoost classifier achieves consistently stronger performance across most metrics, particularly in terms of F1-score and ROC-AUC. This indicates that XGBoost is better able to capture complex demand patterns arising from the interaction of multiple business factors. The improved discrimination capability makes it especially useful for prioritising high-risk or high-opportunity situations.

The use of synthetic data contributes to the consistency of these results, as demand patterns were generated with coherent underlying logic. While real SME data may introduce additional noise, the findings demonstrate that the proposed framework can produce reliable classification outcomes when multi-source data is integrated effectively.

4.4 Global Feature Importance Analysis – Random Forest Models

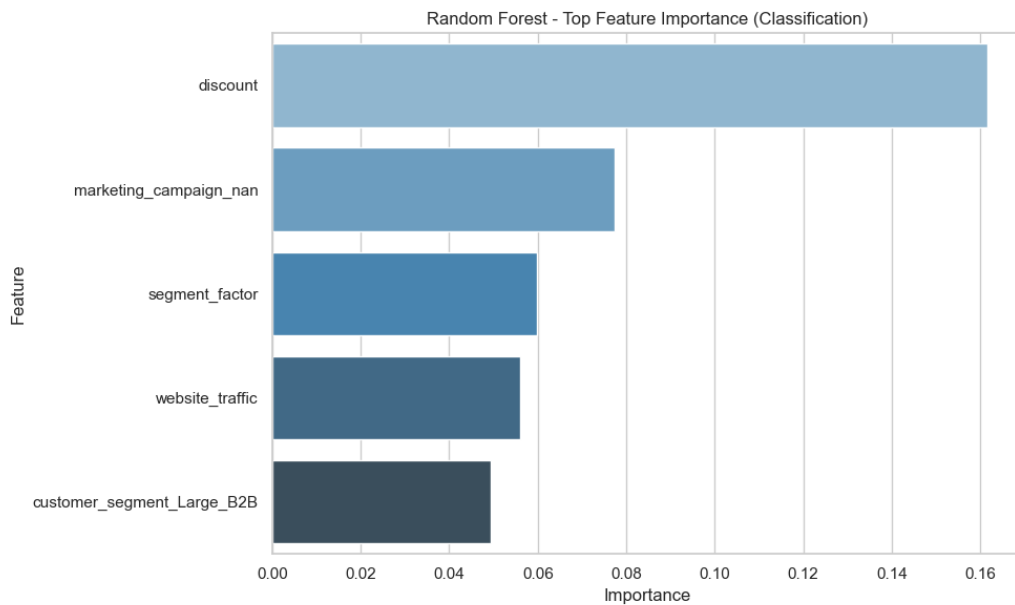
After evaluating predictive performance, the next step is to examine which variables most strongly influence model predictions. Global feature importance analysis provides an overall view of how different factors contribute to revenue and demand outcomes across the entire dataset. For SMEs, this type of analysis is particularly useful because it highlights broad business drivers rather than individual, case-specific explanations.

Random Forest models compute feature importance based on how much each variable contributes to reducing prediction error across multiple decision trees. While this approach does not explicitly model feature interactions, it offers an intuitive and accessible overview of influential variables, making it suitable for an initial interpretation of model behaviour.

FIGURE 1: Random Forest Feature Importance – Regression Model

Source: Own Representation

The regression-based feature importance results indicate that revenue predictions are driven primarily by sales-related variables such as units sold, pricing and discount levels. Marketing activity and seasonal indicators also rank highly, reflecting their direct influence on revenue generation. Operational variables, including inventory levels and logistics-related indicators, appear as secondary drivers, suggesting that operational readiness plays a supporting role in revenue performance. Workforce-related variables contribute more modestly but still provide meaningful explanatory value.

FIGURE 2: Random Forest Feature Importance – Classification Model

Source: Own Representation

For the classification task, which focuses on identifying high-demand situations, the feature importance distribution shifts slightly. Demand classification is influenced more strongly by promotional activity, seasonality and inventory availability. These results align with practical SME decision logic, where demand surges are often linked to marketing campaigns, peak periods and supply-side readiness. External contextual variables also contribute, indicating that demand patterns cannot be explained solely through internal business metrics.

It is important to note that Random Forest feature importance reflects overall influence rather than causal relationships. Some variables may appear less important individually even though they contribute indirectly through interactions with other features. This limitation motivates the use of more advanced models and complementary explainability techniques in subsequent sections.

Overall, the Random Forest feature importance analysis confirms that SME performance outcomes are shaped by a combination of commercial, operational and contextual factors.

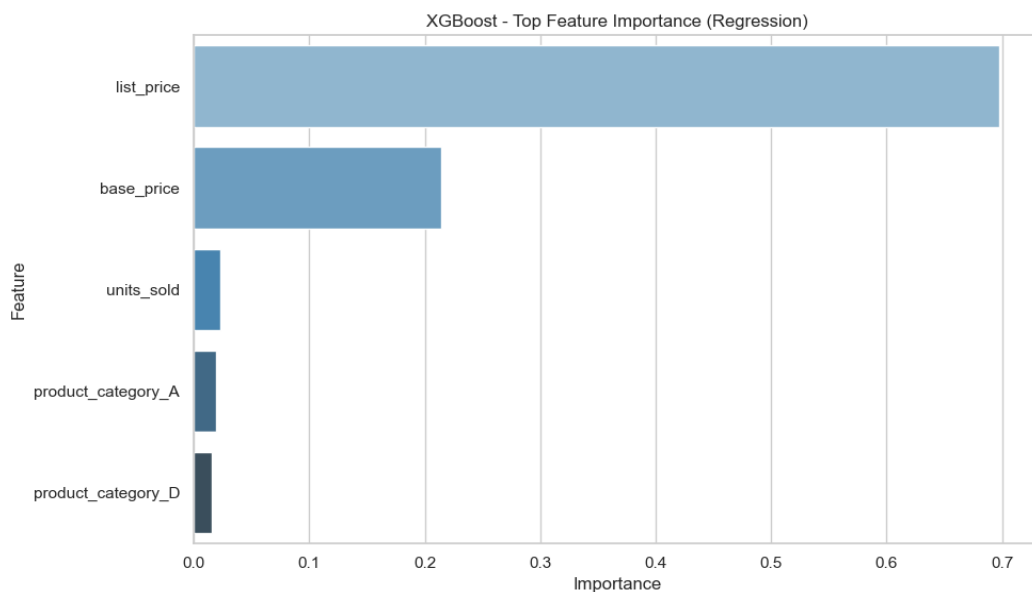
The results provide a useful high-level overview of key drivers and serve as a baseline for deeper interpretability analysis using XGBoost and local explanation methods.

4.5 Global Feature Importance Analysis – XGBoost Models

Following the baseline feature importance analysis using Random Forest, this section examines global feature importance derived from the XGBoost models. XGBoost was selected as the primary predictive model due to its superior performance across both regression and classification tasks. Its gradient boosting structure allows it to capture more complex, non-linear relationships and interaction effects between variables, making its feature importance outputs particularly informative for strategic analysis.

Unlike Random Forest, XGBoost evaluates feature importance based on how frequently and effectively features are used to improve model performance during the boosting process. As a result, the importance rankings often reflect deeper structural relationships within the data rather than surface-level correlations. This makes XGBoost well suited for analysing multi-source SME datasets where outcomes are driven by interacting business factors.

FIGURE 3: XGBoost Feature Importance – Regression Model

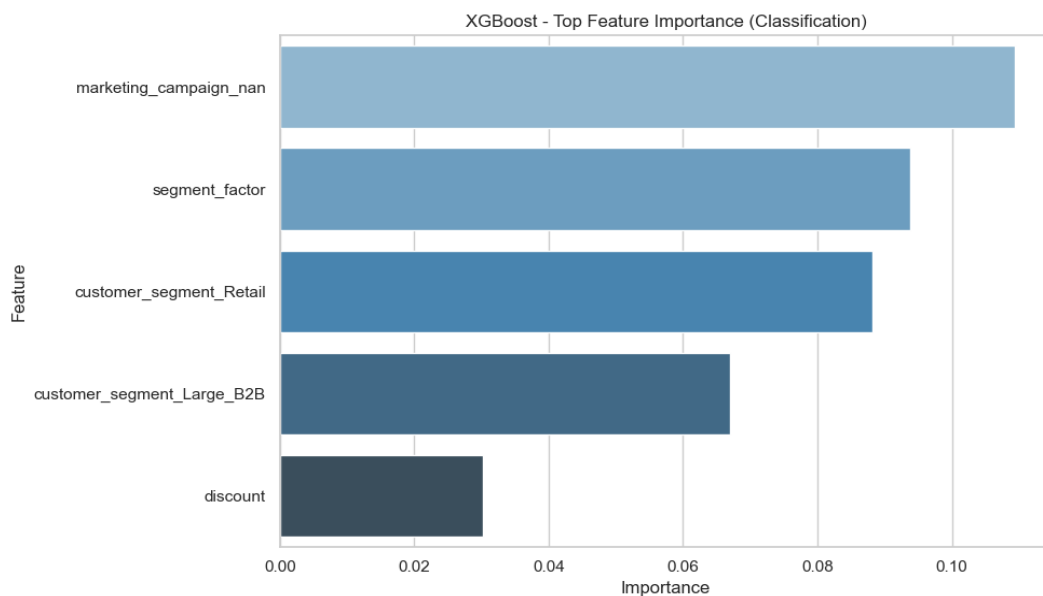


Source: Own Representation

The regression-based XGBoost feature importance results show a more distributed influence across variables compared to the Random Forest model. Revenue predictions are driven not only by direct sales indicators but also by operational efficiency, workforce productivity and external contextual variables. Pricing-related features and marketing intensity continue to play a significant role, but their influence is often moderated by operational constraints such as inventory availability and logistics performance.

For the classification task, which focuses on identifying high-demand situations, XGBoost places greater emphasis on variables related to demand triggers and capacity readiness. Promotional indicators, seasonality effects and inventory-related features rank among the most influential, reflecting their importance in determining whether demand exceeds critical thresholds. External market signals also contribute, reinforcing the value of incorporating contextual data into demand classification models.

FIGURE 4: XGBoost Feature Importance – Classification Model



Source: Own Representation

Compared to Random Forest, the XGBoost classification model demonstrates a clearer separation between primary and secondary demand drivers. This suggests that XGBoost

is better able to prioritise features that are most relevant for identifying high-risk or high-opportunity demand situations, which is particularly valuable for SMEs operating under tight resource constraints.

Overall, the XGBoost feature importance results provide deeper insight into the structure of SME performance drivers. The findings confirm that strategic outcomes are shaped by interactions across sales, operations, workforce and external environments. These results further justify the selection of XGBoost as the core predictive model within the proposed framework and set the stage for the local explainability analysis presented in the next section.

4.6 Local Explainability Analysis Using LIME

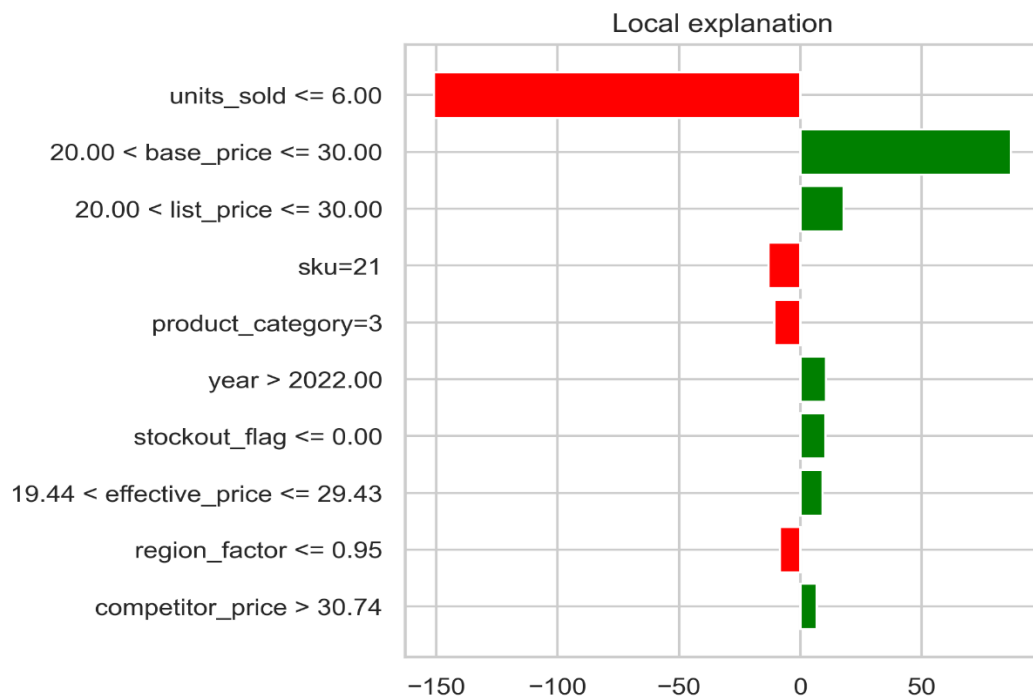
Global feature importance gives a general idea of which variables affect model predictions, but SMEs typically need explanations at the individual case level to make strategic decisions. Most of the time, managers want to know why a certain prediction was made for a certain product, area, or time frame. To meet this need, Local Interpretable Model-agnostic Explanations (LIME) were used to create instance-level explanations for both regression and classification models.

LIME clarifies single predictions by locally approximating the intricate machine learning model with a more straightforward, comprehensible model. This makes it easy to see how each attribute affects a certain prediction. Because LIME works with any model, it can be used with any algorithm, which makes it a good choice for real-world decision-support systems.⁸⁴

⁸⁴ Cf. Ribeiro, M. T., Singh, S. & Guestrin, C., “Why Should I Trust You?” Explaining the Predictions of Any Classifier, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.

4.6.1 LIME Explanations for Revenue Predictions

FIGURE 5: LIME Explanation for Revenue Prediction



Source: Own Representation

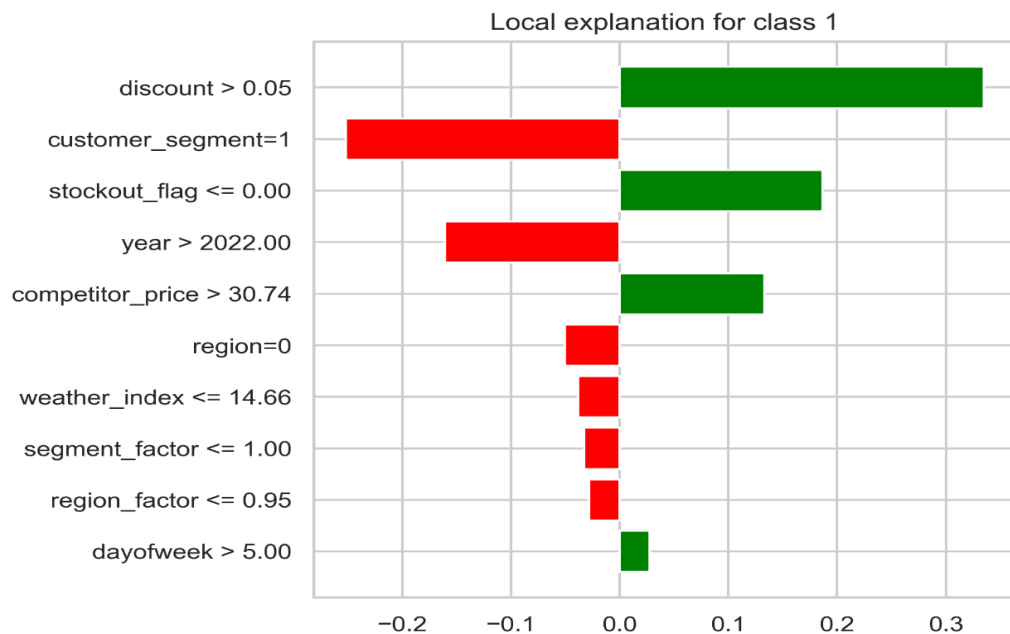
The LIME regression explanation illustrates how individual features contribute to a single revenue prediction. The visualisation shows both positive and negative contributions, indicating which factors increased or reduced the predicted revenue relative to the baseline. Sales-related variables, such as units sold and discount intensity, typically exert strong positive influence, while operational constraints or unfavourable external conditions may reduce the predicted outcome.

From a managerial perspective, this form of explanation is particularly valuable because it aligns closely with real decision-making processes. Rather than presenting abstract importance scores, LIME highlights concrete factors that decision-makers can potentially influence, such as pricing strategies, promotional activity or operational readiness. This enables managers to assess whether the prediction aligns with business intuition and to identify actionable levers for future improvement.

The clarity of the LIME output is enhanced using synthetic data, which ensures consistent relationships between variables. This contributes to stable and interpretable explanations, reducing the risk of contradictory or misleading insights that can arise from noisy real-world data.

4.6.2 LIME Explanations for High-Demand Classification

FIGURE 6: LIME Explanation for High-Demand Classification



Source: Own Representation

The LIME classification explanation shows how each feature affects the model's choice to label an observation as high demand. Promotional indicators, seasonality effects, and inventory availability are all factors that usually make the prediction lean toward a high-demand classification. On the other hand, capacity constraints or weaker market signals may make this effect less strong.

These explanations are especially helpful for small and medium-sized businesses (SMEs) because they help them make quick decisions about operations and strategy. For instance,

managers can use the explanation to figure out if a predicted rise in demand is caused by factors that can be controlled within the company or by outside conditions that cannot be changed right away. This difference is important for choosing whether to grow the business, change marketing strategies, or be careful with risk management.

Overall, the LIME-based explanations show that predictions made by complex machine learning models can be turned into insights that are easy to understand and useful for making decisions. LIME helps build trust, accountability, and informed decision-making by making it clear how each prediction was made. These are all important factors for the adoption of artificial intelligence in small and medium-sized businesses.⁸⁵

⁸⁵ Cf. Schwaeppe, J., *The New Normal: The Status Quo of AI Adoption in SMEs, Technological Forecasting and Social Change*, 2025.

5. Discussion

5.1 Purpose of the Discussion

This chapter is about the outcomes of Chapter 4 and their connection to the study aims and the overall picture of small and medium-sized businesses (SMEs) strategies, data-driven decision-making, and explainable AI. Chapter 4 was mostly about showing actual results. This chapter's goal is to explain why these results are important, how they are relevant, and how they help both academic understanding and the ability of small businesses to make decisions.

The talk is about the main parts of the suggested structure: LIME's ability to make things clear, forecast performance, and the ability to mix data from different places. We look at how these parts work together to help small and medium-sized businesses (SMEs) deal with the problems they have when they want to use AI-based decision-support systems and advanced analytics.⁸⁶

5.2 Interpretation of Predictive Performance Results

The prediction results indicate that the integration of multi-source data into a unified analytical pipeline enables machine learning models to significantly improve strategic decision-making in SME contexts. XGBoost consistently surpassed Random Forest in evaluation metrics, exhibiting exceptional performance in both regression and classification assessments.

XGBoost's better performance shows that gradient boosting methods work well with structured business data, especially when results are affected by complex, non-linear interactions. This finding aligns with prior research illustrating the efficacy of boosting algorithms in capturing interaction effects across various organizational dimensions. This means that complex ensemble models can be very helpful for making predictions in the

⁸⁶ Cf. Hevner, A. R., March, S. T., Park, J. & Ram, S., Design Science in Information Systems Research, MIS Quarterly, Vol. 28, No. 1, 2004, pp. 75–105.

case of small and medium-sized businesses (SMEs), even when the data comes from different places.⁸⁷

When computational resources or technical experience are limited, SMEs may benefit from simpler ensemble approaches, as indicated by the robust baseline performance of Random Forest models. This supports the idea that the effectiveness of AI in SMEs depends on the alignment of analytical tools with organizational skills and decision-making requirements, rather than just model sophistication.⁸⁸

It is also important to consider the role of synthetic data in interpreting these results. The controlled structure of the dataset ensures consistent relationships between variables, contributing to stable and interpretable model behaviour. While real-world SME data may introduce additional variability, the results demonstrate the analytical feasibility of the proposed framework and provide a clear proof of concept for real organisational applications.

TABLE 8: Interpretation of Predictive Model Performance

Model	Predictive Strength	Practical Interpretation for SMEs
Random Forest	Stable baseline accuracy	Suitable where interpretability and low complexity are prioritised
XGBoost	Higher predictive precision	Appropriate for strategic forecasting with complex interactions

Source: Own Representation

⁸⁷ Cf. Chen, T. & Guestrin, C., *XGBoost: A Scalable Tree Boosting System*, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794.

⁸⁸ Cf. Hastie, T., Tibshirani, R. & Friedman, J., *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009, pp. 337–360.

5.3 Value of Multi-Source Data Integration for SME Decisions

One of the main goals of this research was to see if combining different data sources makes strategic decision help for small and medium-sized businesses (SMEs) better and more useful. The results strongly support this assumption. Combining sales, financial, human resource, operational, and external environmental data into one analytical framework made it easier to understand and more accurate at making predictions.

The studies on feature importance show that interactions between different functional areas, not just single factors, affect how well SMEs do. At the same time, price plans, marketing campaigns, employee productivity, inventory levels, and outside factors all affect revenue and demand results. This data backs up the idea that SME decision-making is inherently cross-functional and can't be fully supported by analytics that are only for one area.

From a strategic point of view, integrating data from multiple sources gives decision-makers a better picture of how the business works and lets them look beyond just performance metrics. This makes it easier for different parts of the organization to better anticipate risks, find new opportunities, and align their strategic actions. The results show that multi-source analytics is a strategic need for small and medium-sized businesses (SMEs) to get significant AI-driven decision support, not just a technical upgrade.

TABLE 9: Strategic Contribution of Multi-Source Data to SME Decisions

Data Domain	Key Variables	Strategic Insight Enabled
Sales & Marketing	Promotions, pricing	Seasonality, market signals
Operations	Inventory, lead time	Stockout risk and capacity planning
HR	Productivity, absenteeism	Workforce impact on performance
External	Seasonality, market signals	Context-aware forecasting

Source: Own Synthesis

5.4 Explainability and Managerial Trust

This thesis's demonstration of how explainable AI may combine management usability and prediction accuracy is among its most significant contributions. While local explainability through LIME is particularly helpful for decision-makers in small and medium-sized firms (SMEs), global feature importance provides valuable high-level information.

Chapter 4's LIME explanations demonstrate how to rationally deconstruct specific predictions. By providing a realistic explanation for why a particular outcome was anticipated, this aids managers in thinking. This kind of transparency is essential for acceptance and trust in small and medium-sized enterprises (SMEs), because a limited number of individuals oversee making choices. The results show that business intuition and LIME explanations align well. Seasonality, capacity limitations, inventory availability, and promotions are all presented in a form that makes sense for actual decision-making rather than merely abstract numbers. This supports the notion that explainability, rather than technical completeness, should be determined by how well people can grasp it.

Additionally, explainability aids decision-makers in distinguishing between controllable and uncontrollable factors. This distinction is particularly important for strategic planning because it establishes whether managers should take corrective action or adopt a more cautious stance in the face of anticipated changes. The findings show that explainable AI raises the standard of strategic decision-making as well as trust.

5.5 Implications for SME Strategic Decision-Making

The study's results have a lot of important effects on how SMEs should plan. First, they show that AI-based decision support systems can be both accurate and easy to understand when they are made with SME limitations in mind. This puts to rest the idea that only big companies with advanced data systems and specialized knowledge can use advanced analytics to their advantage.

Second, the analytical workflow's use of explainability makes sure that AI doesn't replace management judgment; it works with it. The framework lets small and medium-sized businesses use AI to help them make decisions by making things clearer, less uncertain,

and more accountable. This backs up the idea that AI needs human-machine cooperation, not full automation, to be used successfully.

Third, the framework stresses how important it is to match analytical tools with strategic goals. The proposed strategy directly addresses critical decisions vital for SME survival and growth by focusing on revenue forecasting, demand categorization, and transparency. This makes it more likely that these technologies will be used and kept up in the real world.

5.6 Contribution to Academic Research

From an academic perspective, this thesis contributes to the literature by integrating multiple research streams that are often examined separately. While existing studies address big data analytics, decision support systems, explainable AI or SME strategy in isolation, this research demonstrates how these elements can be combined into a coherent, end-to-end framework.⁸⁹

The study extends existing work on explainable AI by operationalising LIME within a multi-source SME decision context, rather than treating explainability as a purely technical concern. It also contributes to SME analytics literature by providing a practical framework that balances predictive performance with usability and interpretability.

Furthermore, the use of synthetic data as a methodological tool demonstrates how controlled datasets can be used to evaluate complex decision-support architectures when real organisational data is unavailable. This offers a valuable approach for future research exploring AI applications in sensitive or data-constrained environments.

5.7 Limitations and Reflection

Despite its contributions, this study has several limitations that should be acknowledged. The primary limitation relates to the use of synthetic data. While the dataset was designed to reflect realistic SME conditions, it cannot fully capture the noise, inconsistencies and

⁸⁹ Cf. Gunning, D. et al., XAI—Explainable Artificial Intelligence, Defense Advanced Research Projects Agency (DARPA), 2019.

behavioural complexities present in real-world data. As a result, model performance and explainability may differ when applied to actual SME datasets.

Additionally, the study focuses on a limited set of machine learning models and explainability techniques. Although Random Forest, XGBoost and LIME are widely used and well suited to the research objectives, future work could explore additional algorithms or alternative explainability approaches.

Finally, the evaluation of explainability is primarily qualitative and interpretive. While the explanations appear intuitive and decision-relevant, future research could incorporate user studies to assess how SME managers perceive and utilise explainable AI outputs in practice.

6. Conclusion

6.1 Summary of the Study

This thesis set out to design and evaluate an explainable artificial intelligence framework capable of supporting strategic decision-making in small and medium-sized enterprises through the integration of multi-source big data, predictive analytics, and model interpretability. The primary motivation for this research was the growing recognition that, while AI-driven decision support offers substantial potential benefits for SMEs, adoption remains limited due to challenges related to model complexity, lack of transparency, and low managerial trust.

To address these challenges, the study proposed an end-to-end analytical framework that integrates heterogeneous data sources, applies machine learning models for prediction, and incorporates explainability through Local Interpretable Model-agnostic Explanations (LIME). Using a carefully constructed synthetic dataset reflecting realistic SME conditions, the framework was evaluated across both regression and classification tasks relevant to strategic business decisions, such as revenue forecasting and demand identification.

The results demonstrated that advanced machine learning models can generate accurate and stable predictions when multi-source data is effectively integrated, and that explainability mechanisms significantly enhance the interpretability and managerial relevance of these predictions. Taken together, the findings confirm that explainable AI can function as a practical and trustworthy decision-support tool for SMEs when designed with organisational constraints and decision-making needs in mind.

6.2 Key Findings and Contributions

Several key conclusions emerge from this research. First, the findings confirm that integrating multiple data domains, sales, finance, operations, human resources, and external factors, substantially improves the relevance and richness of predictive insights. SME performance outcomes were shown to be driven by interactions across functional areas

rather than isolated variables, reinforcing the need for holistic, cross-functional analytics in strategic decision-making.

Second, the comparative evaluation of predictive models demonstrated that ensemble methods, particularly gradient boosting, are well suited for structured SME data characterised by non-linear relationships and interaction effects. At the same time, the strong performance of simpler ensemble models highlights that SMEs can still derive meaningful value from less computationally intensive approaches, provided the models are appropriately aligned with organisational capabilities.

Third, and most importantly, the study shows that explainability plays a central role in bridging the gap between predictive accuracy and practical usability. This supports informed judgement, enhances trust, and allows decision-makers to distinguish between controllable and uncontrollable drivers of outcomes, an essential requirement for strategic planning in SME contexts.

From an academic perspective, this thesis contributes by integrating multiple research streams, big data analytics, decision support systems, explainable AI, and SME strategy, into a single coherent framework.

6.3 Practical Implications for SMEs

The findings of this study have clear practical relevance for SMEs considering the adoption of AI-driven decision-support systems. The proposed framework demonstrates that advanced analytics does not need to remain the exclusive domain of large organisations with extensive data infrastructure. By combining interpretable models, explainability techniques, and integrated data pipelines, SMEs can adopt AI in a way that complements managerial expertise rather than replacing it.

The framework provides SMEs with a blueprint for using AI as a decision aid that enhances understanding, reduces uncertainty, and supports accountability. By embedding explainability directly into the analytical workflow, decision-makers can engage with model outputs confidently and critically, increasing the likelihood of sustained adoption. Furthermore, the use of lightweight, model-agnostic explanation techniques such as

LIME reduces the technical burden associated with deploying explainable AI, making it a realistic option for resource-constrained organisations.

6.4 Limitations of the Study

Despite its contributions, this research has several limitations that should be acknowledged. The primary limitation concerns the use of synthetic data. Although the dataset was designed to reflect realistic SME environments and cross-functional interactions, it cannot fully capture the complexity, noise, and behavioural dynamics present in real organisational data. As a result, model performance and explanation stability may vary when applied to real-world SME datasets.

Additionally, the study focuses on a limited set of machine learning models and a single explainability technique. While Random Forest, XGBoost, and LIME are widely adopted and well suited to the research objectives, alternative models or explanation methods may offer different trade-offs between accuracy, interpretability, and computational cost.

Finally, the evaluation of explainability is primarily analytical and interpretive. Although the explanations generated appear intuitive and decision-relevant, the study does not include empirical user validation involving SME managers. As such, conclusions regarding trust and usability are based on logical alignment with managerial decision processes rather than direct behavioural evidence.

6.5 Directions for Future Research

Future research could extend this work in several important directions. First, applying the proposed framework to real SME datasets would allow for validation under practical constraints such as data quality issues, organisational heterogeneity, and evolving business conditions. Such studies would strengthen external validity and provide deeper insight into real-world adoption challenges.

Second, future studies could explore alternative or complementary explainability techniques, including global explanation methods or hybrid approaches that combine local and global interpretability. Comparing different XAI methods within SME decision

contexts would further clarify which approaches best support managerial understanding and trust.

Third, incorporating qualitative user studies or controlled experiments involving SME decision-makers would provide valuable evidence on how explainable AI influences decision behaviour, confidence, and accountability in practice. This would enable a more rigorous assessment of explainability beyond technical correctness, focusing instead on human AI collaboration outcomes.

6.6 Final Remarks

In conclusion, this thesis demonstrates that explainable artificial intelligence can play a meaningful role in enabling data-driven strategic decision-making in small and medium-sized enterprises. By integrating multi-source data, predictive modelling, and transparent explanations into a single framework, the study shows that AI can be both powerful and accessible when designed with human decision-makers in mind.

The proposed framework offers a practical pathway for SMEs seeking to leverage AI responsibly, transparently, and effectively. As organisations increasingly rely on data-driven insights to navigate competitive and uncertain environments, the ability to understand and trust AI-driven recommendations will become ever more critical. This research contributes to that objective by showing how explainability can transform complex machine learning outputs into actionable strategic insight.

Appendices

Appendix A: Dataset Documentation

TABLE 10: Overview of Dataset

Attribute	Description
Application Domain	SME Strategic Decision Making
Data Structure	Tabular
Data Types	Numerical and Categorical
Data Sources	Sales, Finance, HR, Operations, External Indicators
Number of Records	40,000
Number of features	50

Source: Own Representation

TABLE 11: Target Variables

Target Variable	Data Type	Task	Relevance
Revenue	Continuous	Regression	Finance Forecasting
High Demand Flag	Binary	Classification	Demand Planning

Source: Own Representation

Appendix B: Python Source Code

The complete Python source code used for data generation, data preprocessing, model training and analysis in this study has been submitted as a supplementary digital file.

- File Name: Thesis_749486.ipynb
- Format: Jupyter Notebook
- Language: Python
- Reproducibility: Fully reproducible

TABLE 12: Functional Scope of Codebase

Component	Function
Data pipeline	Synthetic Data Generation and preprocessing
Feature Engineering	Encoding and Transformation
Model Training	Random Forest and XGBoost
Model Evaluation	Performance Metrics
Explainability	Global feature importance

Source: Own Representation

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Declaration in lieu of oath

I, Anurag Akshay Rathore, solemnly declared that the report titled " An Explainable AI Framework for Multi-Source Big Data in SME Strategic Decision-Making" is entirely my own work. I confirm that all the ideas, theories, opinions, and information presented within this report are original unless appropriately cited and referenced. I have diligently acknowledged all sources consulted during the preparation of this report to avoid any instances of plagiarism or academic dishonesty. Furthermore, I assert that I have not previously submitted this report, or any similar version of it, for academic credit elsewhere. I fully understand the gravity of academic misconduct and accept full responsibility for the authenticity and integrity of this work. By affixing my signature below, I pledge my commitment to upholding the principles of academic honesty and excellence.

Essen, 03/01/2026

Anurag Akshay Rathore

Place, Date

Signature